

Havlík’s Law as a Syllabic Rule: A Structural Reanalysis of the Slavic Jers

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Abstract

Havlík’s Law is the classical description of jer behavior in Common Slavic, but its diachronic counting rule cannot be the synchronic rule modern Slavic speakers apply productively to new lexemes. This paper argues that Havlík’s Law is not a counting rule at all but the surface signature of a syllabification mechanism that operates at both temporal slices. A small ranked-constraint syllabifier decides which orthographic positions function as prosodic nuclei. Jers are not nuclei by default; they are promoted only when no other segment can serve.¹

The central empirical contribution is two strictly sealed held-out tests totalling 60 OCS forms selected to lie outside every earlier corpus, with expected masks hand-labeled before each parser run, parameters frozen and md5-recorded, and the parser executed exactly once per sealed corpus. **Combined result: 58/60; the two failures correspond to documented inventory gaps anticipated in advance.** The held-out test is supported by three further results: design corpus consistency on the Havlík corpus and the modern Serbian *nepostojano* A paradigmatic alternation (95 of 95 in-scope across three pre-design corpora); a non-stipulated structural prediction (a monosyllabic/polysyllabic split for word-initial sonorant forms recovering the minimal-word effect without an explicit constraint, corroborated by Serbian *rda*, *rdav*, *rvati*, *mrknuti*); and a cross-branch statistical signature on $n = 29 + 30$ across six Slavic branches showing categorically more divergence at jer-source positions than real-vowel positions (Mann-Whitney $U = 44$, $z = -5.921$, $p < 10^{-6}$, rank-biserial effect = 0.899). A second, qualitatively different result complements this: single-parameter changes to the OCS-default parser reproduce daughter-branch surface forms (Serbian *danas* from disallowing the /dn-/ onset; Russian волк from penalizing syllabic /l/). Both are instances of one principle—when phonotactic pressure builds, the slot fills—and neither corresponds to any prediction the counting rule itself can make.

The framework—empty-nucleus syllabification with ranked constraints—is not itself novel, aligning with Government Phonology (Kaye, 1990; Lowenstamm, 1996; Scheer, 2004, 2005) and classical generative phonology (Lightner, 1965; Gussmann, 1980; Rubach, 1984). What is novel is the unification: Havlík’s Law, modern Slavic *nepostojano* A, and a structurally tight sub-class of word-initial syllabic resonants reduce to a single mechanism, mechanized as an auditable algorithm and validated

*Companion code, parameters, and test corpora are openly available at <https://github.com/dzeremy/syllabic-havlik>.

under sealed-test methodological discipline. The diachronic claim we defend is restricted: the jers were not vocalic *at the time of the alternation pattern that became Havlík’s Law* (approximately the 9th century onward); earlier phonetic content is treated as a separate question (Section 10.2.1).

1 Havlík’s Law and what it does not explain

A note on periodization. We use the following terms throughout: *Proto-Slavic* for the earliest reconstructible stage; *Common Slavic* for the reconstructed proto-language broadly; *Late Common Slavic* (approximately the 7th–9th centuries CE) for the immediately pre-breakup period in which the strong/weak distribution stabilized; and *Old Church Slavonic (OCS)* for the orthographic tradition attested from the 9th–11th centuries CE. When chronological precision matters, we specify the period directly (“the 9th century onward,” etc.). The slot view’s central diachronic claim, as foregrounded in the abstract and defended in Section 10.2.1, applies to Late Common Slavic and the OCS attestation; earlier Proto-Slavic stages may have had different phonetic content and are treated as a separate question.

Havlík’s Law (Havlík, 1889) is one of the few named sound laws of Slavic historical phonology that every introductory grammar reproduces. Its content, in modern textbook formulation (Shevelov, 1964; Carlton, 1991; Lunt, 2001), is that in a sequence of jers (the ultra-short vowels traditionally written ъ and ѣ in OCS), every other jer counting from the right edge alternates between two states: the rightmost is *weak* and deletes; the next is *strong* and vocalizes to a full vowel; the next is weak again; and so on. The vowel quality of strong jers is held to track the underlying jer’s frontness or backness: front ъ vocalizes to /e/ in Czech, /a/ in Serbian, /e/ in Russian; back ѣ vocalizes to /e/ in Czech, /a/ in Serbian, /o/ in Russian.

The law’s empirical reach is genuinely impressive. It predicts that OCS **sъnъ* ‘dream’ yields Czech *sen*, Serbian *san*, Russian сон—the final ѣ being weak and lost, the medial ѣ being strong and surfacing. It predicts that **dъnъsъ* ‘today’ yields Czech *dnes* via three jers in alternating strong-weak-strong (counting from the right edge) classification, the middle jer vocalizing and the outer two disappearing. Across the OCS lexicon the law correctly classifies the overwhelming majority of jer positions in a way no analyst could match by hand inspection.

What the law does not do is **explain** any of this. The counting rule is a description, not a mechanism. It says *which* jers surface as vowels in modern reflexes; it gives no account of *why* every-other-from-the-right should be the relevant computation, or why some jers in the count produce unexpected behavior, or why a sub-class of Old Czech and Old Polish forms shows what Scheer (2005) terms “non-etymological jers”—positions where a vowel surfaces despite the original form having no orthographic jer at

all (Lamprecht, 1987; Komárek, 1969).

1.1 Three things the counting rule does not handle

We pause to identify three specific empirical phenomena that any analyst of the jer alternation eventually has to address, and which the counting rule either fails to predict or covers by stipulation.

First, the Category B liquid-context asymmetry. OCS forms with strong-position jers adjacent to a liquid—words like **vblkǫ* ‘wolf’, **tbrnǫ* ‘thorn’, **pblnǫ* ‘full’—do not vocalize the jer in the predicted way. The companion paper (Jevremović, 2026a) documents this in detail: across Czech, Serbian, and Slovak, 0 of 27 liquid-adjacent strong jers vocalize, while 68 of 68 non-liquid-context strong jers do. The asymmetry is categorical ($p < 10^{-14}$), and Russian’s apparent counterexamples (волк, смерть) produce vowel quality conditioned by the liquid, not by the jer (Townsend and Janda, 1996). The Russian pattern is sharper still on closer inspection: the front jer ь in liquid-adjacent position gives /e/ before /r/ (верх from **vbrxǫ*, зерно from **zbrno*, смерть from **sǫmьrtǫ*) but /o/ before dark /l/ (волк from **vblkǫ*, полный from **pblnǫ*, жёлтый from **žbltǫ*). Under Havlík’s law a front jer should vocalize uniformly to /e/ regardless of the following liquid. The observed /e/-versus-/o/ split is amenable to multiple accounts: the standard reading handles it via the well-known Russian-internal /e/ → /o/ backing before non-palatalized consonants (Townsend and Janda, 1996), in which the front jer first vocalizes to /e/ uniformly and then backs to /o/ before dark /l/ via the independent backing rule. The slot view derives the same pattern without the intervening Havlík step: the liquid-conditioned context delivers /o/ directly, no jer-internal feature required. Both accounts produce the correct surface; the slot view’s appeal here is parsimony (one mechanism instead of two) and the unification with the cross-branch divergence result (Section 7) where the per-branch resolution varies more sharply than a single proto-quality can support (Jevremović, 2026a). The Russian pattern is suggestive support rather than knockout disconfirmation of the alternative.

Second, the synchronic paradigmatic alternation. In modern Serbian, the lexeme *san* ‘dream’ has genitive *sna*; the lexeme *borac* ‘fighter’ has genitive *borca*; the masculine adjective *mrtav* ‘dead’ has feminine *mrtva*. The /a/ that appears in one form disappears in another. Standard Serbian grammars (Belić, 1962; Stanojčić and Popović, 1992; Klajn, 2005) treat this collectively as **nepostojano A** (‘unstable A’), and the phenomenon is *productive*: a borrowed word inflects with nepostojano A if it has the right structural shape, with no historical jer source available to license the alternation.

The issue here is not that Havlík’s Law produces wrong surface outputs for forms like *san/sna*—it does not; the diachronic derivation gives the right answers. The issue is that Havlík’s Law cannot *be* the rule that modern Serbian speakers apply when they

inflect new words. Havlík’s Law is a description of a one-time historical event—the fall of the jers, completed by approximately the 12th century. A 9th–12th-century historical event cannot be the synchronic rule a 21st-century speaker applies to a loanword that entered the language last year. The standard Havlík account therefore leaves a gap: what *is* the modern synchronic rule that produces the nepostojano A alternation, and why does it have exactly the shape Havlík describes diachronically? The standard description has no answer beyond “this is what the grammar happens to do.” A theory that unifies the diachronic and synchronic regularities under a single mechanism explains the productivity automatically; the slot view does this, and Havlík does not.

Third, a sub-class of OCS forms with word-initial syllabic resonants. For forms like **rǫděti* ‘to redden’, **rǫvati* ‘to tear’, and **mǫrgnǫti* ‘to wink’, Havlík predicts deletion of the single jer (no other jer in the chain). The predicted output for **rǫděti* is **/rděti/*. But */rd-/* is not a legal Slavic onset under any conservative inventory: sonority falls from */r/* to */d/*, violating the cross-linguistic Sonority Sequencing Principle. The predicted form requires either an unattested OCS onset cluster or—tacitly—the analysis that */r/* is syllabic. Modern Serbian preserves the syllabic resonant explicitly in *rđā* ‘rust’, *rđav* ‘bad/rusty’, *rvati se* ‘to wrestle’, *mrknuti* ‘to blink’. The standard reading thus describes the phonological behavior without coherently predicting it.

1.2 The proposal

We propose that Havlík’s Law is not a counting rule at all. It is the surface signature of a syllabification mechanism. The mechanism is straightforwardly stated:

Every syllable requires a nucleus. Real vowels (those that are stable across paradigms) are pinned as nuclei. Other segments—orthographic jers and liquids between non-vowels—may serve as nuclei when structural pressure requires. A constraint set ranks the possible nucleus assignments, and the lowest-scoring legal syllabification wins.

The rule has no jer-counting; the apparent counting effect emerges from the constraint ranking. The rule has no notion of “strong” or “weak” jers; those concepts are post-hoc descriptions of the structural classification the ranking produces. The substantive analytical content of the rule is concentrated in a single constraint, foregrounded in Section 3.2 below: **a jer cannot function as the nucleus of an open medial syllable**. This is the rule that determines when a slot fills (surfaces as a vowel) and when it stays empty; the rest of the constraint set provides the supporting machinery. With this constraint in place, the rule generalizes naturally to the three problems above: the Category B asymmetry falls out because liquids are preferred over jers as syllabic nuclei when both are available; nepostojano A falls out because the same rule applied

to modern Serbian paradigms produces vowel-zero alternation conditioned on the rest of the prosodic word; and the trilemma sub-class falls out because the rule forces the syllabic-resonant analysis exactly where standard Havlík makes incoherent predictions.

The view defended here aligns with empty-nucleus analyses in Government Phonology and CVCV phonology (Kaye, 1990; Lowenstamm, 1996; Scheer, 2004, 2005) and with the abstract-vowel tradition in classical generative phonology (Lightner, 1965; Gussmann, 1980; Rubach, 1984). Scheer (2005) in particular argues explicitly that the objects we call jers are not vowels but empty nuclei filled by structural pressure. We adopt that position substantively. What we add is (a) a constraint-based mechanization auditable as an algorithm, (b) validation across three test corpora at OCS, synchronic-Slavic, and word-initial-sonorant levels, and (c) the structural argument from the trilemma sub-class, which to our knowledge has not been previously assembled.

2 The structural reformulation

We restate Havlík’s Law as a syllabification rule and identify the prior work whose framework we adopt.

2.1 From counting to ranking

The traditional formulation of Havlík’s Law is a counting rule: alternate strong and weak from the right edge of the jer chain. The strong jers surface, the weak jers delete. The counting is presented as a primitive operation, with no further phonological motivation. The most common move in subsequent literature (Lightner, 1965; Gussmann, 1980; Rubach, 1984, 1986) has been to derive the counting by positing an abstract underlying vowel in every jer position, with a deletion rule conditioned on the structural context. The deletion rule is itself a counting rule, restated in autosegmental terms.

The structural reformulation eliminates the counting entirely. The mechanism is syllabification: given a CS input string, the syllabifier identifies candidate nuclei (real vowels, jers, and liquids between non-vowels), enumerates possible nucleus assignments, ranks each by a small set of constraints, and returns the lowest-scoring legal parse. A jer that surfaces as a vowel in the winning parse is what the standard description calls “strong”; a jer that does not is what it calls “weak.” The apparent alternating pattern is a consequence of the constraints, not the input.

The key analytical move is that jers are not vowels at the lexical level. They are **structural slots**—empty prosodic nuclei in the sense of Scheer (2004)—that the syllabifier may either fill or leave empty depending on what else is in the prosodic word. The slot fills when no real vowel can anchor the syllable; it stays empty when one can. The standard “strong” jers are filled slots; the standard “weak” jers are slots that remain

empty.

2.2 Relation to prior work

The structural reformulation is not novel as a theoretical proposal. Its closest published antecedents are:

- [Lightner \(1965\)](#), working on Polish, posits abstract underlying jers deleted by a rule conditioned on structural context. The substantive predictions for modern Slavic alternation largely overlap with ours.
- [Gussmann \(1980\)](#) and [Rubach \(1984, 1986\)](#) develop the abstract-vowel account within autosegmental and lexical phonology frameworks. The mechanism is still counting-equivalent, but the formal apparatus is structural.
- [Kaye \(1990\)](#) and [Lowenstamm \(1996\)](#) reformulate the abstract vowel as an empty nucleus within Government Phonology, with the deletion rule reinterpreted as lateral government from a following filled nucleus. The phonological object is now structural rather than featural.
- [Scheer \(2004, 2005\)](#) develop this into the CVCV framework, where syllable structure is a strict sequence of empty CV slots that fill or remain phonetically null. [Scheer \(2005\)](#) explicitly proposes that “objects that undergo vocalization are not yers...the items that undergo vocalization are ‘nothing,’ or zeros. Zeros of vocalic nature, to be precise. In an autosegmental framework, a ‘vocalic zero’ can be nothing else than an empty nucleus.”
- [Koenig \(2004\)](#), working independently in the autosegmental tradition (Rubach, Gussmann) without engaging Government Phonology or CVCV, arrives at a structurally similar conclusion: a jer is “an unlinked feature melody and skeletal slot” (p. 99–100), where the skeletal X-slot exists prior to vocalization but is invisible to the syllable parser because the melodic features are not linked to it. Vocalization (her YVoc rule) fires “when followed by an unsyllabified consonant”—a structural condition, not a counting one—and syllabification proceeds right-to-left. The two empty-nucleus traditions (Government Phonology / CVCV and autosegmental phonology) converge on the same conclusion from different starting points. Where [Koenig \(2004\)](#) differs substantively from the position adopted here is in pre-specifying the melodic features (the actual vowel content) on the unlinked melodic tier; on that account, the surface filling vowel is recovered from the pre-specified melody. The slot view adopted in this paper has only a [+front]/[+back] feature on the slot, with the surface filling vowel determined by a branch-specific resolution rule. This difference is empirically testable through cross-branch divergence

(Section 7), where the slot view predicts the categorical variation that Koenig (2004)’s account does not derive.

We adopt the Scheer (2005) position substantively. What this paper adds is not a new theoretical framework but the demonstration that a small constraint-based syllabifier built on this framework reproduces (i) the Havlík strong/weak distribution across the OCS lexicon, (ii) the categorical liquid-context asymmetry of (Jevremović, 2026a), (iii) the synchronic Serbian nepostojano A paradigmatic alternation, and (iv) a sub-class of word-initial syllabic resonants previously undocumented as a structural prediction. The contribution is empirical and methodological: a mechanism, a corpus-scale validation, and a structural argument from a sub-class of forms where alternative analyses fail.

2.3 What the structural reformulation commits to

The structural reformulation makes the following claims:

1. Jers are not vowels at the Common Slavic lexical level. They are structural slots: empty prosodic nuclei.
2. Slots surface with a vowel when phonotactic pressure forces a nucleus at the position; they stay empty otherwise. The decision is structural, not lexical.
3. Each modern Slavic branch has a branch-specific resolution rule for filling slots, distinguishing front jer (ь) from back jer (ѣ) to varying degrees:

Branch	Front jer (ь)	Back jer (ѣ)
Serbian	/a/	/a/
Czech	/e/	/e/
Polish	/e/	/e/
Bulgarian	/e/	/ə/
Russian	/e/	/o/
Slovak	/e/	mixed (/e/, /a/, /o/)

Serbian, Czech, and Polish collapse the front/back distinction in surface output (one vowel for both jer types); Bulgarian and Russian preserve the distinction at the surface (two vowels, one per jer type); Slovak is mixed, with different resolutions in different contexts. The *structural decision* (slot fills or stays empty) is shared across all six branches; the *filling vowel*, when a slot fills, is branch-specific.

4. Real vowels exist and are diagnosed jointly by orthographic category and paradigmatic behavior, in that order. The OCS scribal tradition distinguishes graphically between full vowels (*a, e, i, o, u, ě, ě, ѡ, y*) and jers (*ь, ѣ*); this graphic distinction predates any modern paradigmatic data and serves as the input criterion for the parser’s pinned-versus-free classification. Paradigmatic alternation in modern reflexes then serves as the diagnostic that confirms whether a given OCS jer position behaves as a slot in modern phonology. The two criteria are independent: the orthographic classification is fixed by 9th-century scribal practice; the paradigmatic diagnostic is a 21st-century test against modern speaker behavior.

The classification is not circular

One could reasonably worry that diagnosing slots through paradigmatic alternation is circular: “what is the parser predicting if the position’s identity is itself defined by the prediction’s target?” The defense rests on two anchors that establish the classification independently of the alternation prediction.

Anchor 1: orthographic input. The pinned-versus-free classification is determined by what the OCS scribe wrote on the page: a vowel grapheme is pinned, a jer grapheme is free. This is an input criterion, not a derived one, and it predates any modern paradigmatic data the parser is being validated against. The parser does not decide which positions are jers; the OCS orthography decides that, and the parser then predicts which free positions surface as nuclei.

Anchor 2: cross-context asymmetry within the same branch. The companion paper (Jevremović, 2026a) documents that the same orthographic jer (ь or ѣ) shows categorically different resolution depending on adjacent structural context. Table 1 summarizes the pattern in the three branches with the clearest contrast.

Branch	Non-liquid context	Liquid context
Czech	/e/: <i>den, pes</i>	syllabic /r/, /l/: <i>vlk, trn</i>
Russian	/e/: <i>денъ</i> , before /r/ <i>сѣрѣх</i>	before dark /l/: /o/, <i>солѣк</i>
Serbian	/a/: <i>san, pas</i>	syllabic /r/ and /l/, later /l/→/u/: <i>vlk</i> → <i>vuk</i>

Table 1: Cross-context asymmetry: the same orthographic jer yields categorically different outputs in non-liquid vs. liquid contexts within each branch. The Russian row contrasts two adjacent-consonant environments for ь (before /r/ vs. before dark /l/), both attested in the same orthographic input.

Within each branch, identical orthographic inputs produce categorically different outputs depending on adjacent context.

A purely outcome-based diagnostic (“slot = position that alternates”) cannot produce this asymmetry: it would classify the same orthographic jer differently depending on its context, but provide no theoretical basis for predicting why. The structural-context dependence is what the slot view explains and what the deletable-real-vowel reading

struggles with. The asymmetry is therefore independent evidence that the slot-versus-real-vowel distinction is a structurally predictive classification, not a post-hoc tautology over outcomes.

The reformulation does **not** claim that all surface vowels in modern Slavic are slot-fillings. The structural rule *does* replace Havlík’s Law as a descriptive matter in the cases where Havlík fails: liquid-context jers, where the orthographic jer does not vocalize at all but marks a syllabic resonant on the adjacent liquid, and the trilemma sub-class (Section 6), where Havlík’s counting prediction produces a phonotactically illegal residue. In the non-liquid Category A cases that Havlík was originally formulated to cover, Havlík describes the surface distribution correctly and the structural rule explains why the description has that shape; in the liquid-context and trilemma cases, Havlík’s surface prediction is wrong and the structural rule supplies the correct one. Nor does the reformulation rule out alternative analyses on a per-form basis. For many non-liquid forms the abstract-underlying-vowel reading and the slot reading produce the same surface predictions; we adopt the structural reformulation on grounds of (a) parsimony, (b) automatic coverage of the three empirical problems identified in Section 1.1, and (c) the structural argument from the trilemma sub-class developed in Section 6, which the alternative reading cannot accommodate without smuggling in the structural analysis itself.

3 A constraint-based syllabifier

We mechanize the structural reformulation as a constraint-based syllabifier. The algorithm:

1. Identify candidate nuclei in the input string. Real vowels are *pinned* as nuclei; jers and liquids between non-vowels are *free* candidates that may or may not surface as nuclei.
2. Enumerate all possible nucleus assignments over the free candidates.
3. For each assignment, enumerate all legal syllabifications by assigning consonants between adjacent nuclei according to a declared onset and coda inventory.
4. Score each syllabification by ranked soft constraints.
5. Return the lowest-scoring legal syllabification as the predicted prosodic structure.

The distinction between pinned and free candidates encodes the distinction between real vowels and slots. A jer assigned the value 1 in the winning parse is a filled slot; a jer assigned 0 is an empty slot.

3.1 The constraint set

The syllabifier uses ten constraints (Table 2). Three are hard (any violation rules out a parse); the rest are soft (violations accumulate as a weighted sum, and the parse with the lowest total wins). One of the soft constraints, *SYLLABIC-RESONANT, carries weight 0 at the OCS ranking. This is not a placeholder: it is a substantive analytical commitment. OCS does not penalize syllabic sonorants, and the parser therefore does not penalize them either. When the hard PARSE constraint requires that every segment belong to some syllable, and the only candidate nucleus in that syllable is a sonorant, the parser builds the sonorant as a syllabic nucleus at zero cost. This is the mechanical instantiation of the trilemma conclusion developed in Section 6: OCS admits syllabic sonorants as a structural class, and the parser predicts them as the optimal output whenever syllabification demands a nucleus and no vowel is available. Which sonorants actually surface as nuclei is then determined by constraint interaction, not by membership in the permitted class; in practice only /r/ and /l/ surface, as discussed below.

Table 2: Constraints used by the syllabifier.

#	Constraint	Type	Description
1	PARSE	hard	Every segment is in some syllable.
2	ONE-NUC-PER-SYL	hard	Exactly one nucleus per syllable.
3	ONSET-LEGAL	hard	Onsets must be in the declared inventory.
4	*JER-NUC-OPEN	10	Penalty for a jer in an open medial syllable.
5	*ILLEGAL-CODA	10	Penalty per coda not in declared inventory.
6	*BARE-NUC	5	Penalty for a syllable with no onset (non-edge).
7	*COMPLEX-CODA	2	Penalty per coda consonant beyond one.
8	*JER-NUC	1	Penalty for any jer-as-nucleus.
9	ONSET-MAX-PREF	0.5	Penalty per coda consonant: maximal-onset tiebreaker.
10	*SYLLABIC-RESONANT	0	Syllabic resonants permitted in OCS; see §6.

Onset and coda inventories are declared from descriptive grammars of OCS and Late Common Slavic (Lunt, 2001; Vaillant, 1948; Carlton, 1991). Illegal codas are treated as soft-penalized rather than hard rejected, reflecting the asymmetric phonotactic profile of Late Common Slavic (Bethin, 1998): highly permissive in onsets, restrictive but not impossible in codas. Constraint *JER-NUC at weight 1 encodes the slot view’s default expectation that jers prefer not to be nuclei—a small bias that breaks ties in favor of non-jer nuclei when both are otherwise legal. Constraint *SYLLABIC-RESONANT at weight 0 records the OCS-level fact that syllabic sonorants are permitted; the parser thereby predicts them as emergent outputs of syllabification rather than stipulating them. Section 9.1 reports a minimal demonstration that raising this constraint to a non-zero weight against an otherwise unchanged OCS parser already produces the Russian волк pattern from **vlkz*, in contrast to the Czech *vlk* pattern that the OCS-default

parser produces. The full per-branch implementation is left to follow-up work.

Predicted absence of syllabic /m/ and /n/. Although *SYLLABIC-RESONANT is at weight 0 for every sonorant alike, the parser’s outputs across all five test corpora contain no syllabic /m/ or /n/. Every syllabic sonorant the parser ever builds is /r/ or /l/. Forms containing /m/ or /n/ in jer-adjacent positions (**gr̥m̥*, **k̥rm̥*, **x̥lm̥*, **č̥ln̥*, **str̥m̥*) are parsed with the nasal as a coda of the adjacent syllable, never as a nucleus. This is not stipulated. It emerges from the interaction of *BARE-NUC (weight 5) with the syllabification preference for hosting consonants in the nearest available onset or coda: making /m/ or /n/ a nucleus would require an additional onsetless syllable, which is more expensive than letting it serve as a coda of an existing syllable nucleated by /r/, /l/, or a filled jer slot. The predicted absence matches the Slavic typology: Czech, Serbian, and Macedonian preserve /r̥/ and /l̥/ as productive nuclei, but no Slavic branch has syllabic /m/ or /n/ in the same way. The parser produces exactly the inventory of syllabic sonorants attested in the daughter branches without explicitly restricting the inventory.

3.2 The central constraint

The constraint set contains ten items, but the analytical work is concentrated in one of them: *JER-NUC-OPEN, at weight 10. This is the constraint that determines, in case after case, whether a slot fills or stays empty. It is, in effect, the formal statement of the slot view’s central insight, and it deserves to be foregrounded.

The constraint says: **a jer cannot function as the nucleus of an open medial syllable**. “Open” here means without a coda; “medial” excludes word-final syllables (handled separately by extrametricality of word-final jers, a Late Common Slavic edge effect). What the constraint expresses, at a deeper level, is a *prosodic anchoring* principle: filled slots require structural support from coda material, and an open medial syllable provides no such support. This is the same principle that drives the typology of vowel epenthesis sites in Itô (1989)’s prosodic theory of epenthesis, the coda licensing patterns in Kaye (1990), and the metrical-grid constraints on weak-vowel licensing in Hayes (1995). A closed syllable provides prosodic anchoring through its coda; an open medial syllable provides none, leaving any filled slot unsupported. The descriptive form of the constraint (“no jer in open medial syllable nucleus”) is the surface signature of this anchoring requirement applied to a slot-filled nucleus.

We adopt the descriptive formulation in the parser because it is mechanically explicit and one-to-one with the structural fact it encodes. The principled formulation (prosodic anchoring) is what justifies the constraint, not a tailored fit; the descriptive name labels its action.

This is what drives the asymmetry between **sɔnɔ* (where the medial ɔ fills the slot because the closed syllable CjC can host it) and **sɔna* (where filling the slot would require either an open medial Cj. syllable or a downstream bare-nucleus violation; the slot stays empty and /sn-/ becomes the onset of a single syllable anchored by /a/). It is what drives the categorical alternation in modern Serbian nepostojano A: *san* versus *sna*, *borac* versus *borca*, *otac* versus *oca*. In every case the form with the filled slot has the slot in a closed syllable; the form with the empty slot would have required an open-syllable filling and so the slot stays empty.

The other constraints in Table 2 are supporting machinery. PARSE, ONE-NUC-PER-SYL, and ONSET-LEGAL establish the basic well-formedness of any syllable structure. *ILLEGAL-CODA captures the Late Common Slavic asymmetric tolerance for cluster types. *BARE-NUC is a structural cost for nucleus-only syllables and is what tips the monosyllabic/polysyllabic split for word-initial sonorants (Section 4.3). *COMPLEX-CODA is a small cost for cluster codas; ONSET-MAX-PREF is a tiebreaker. The bias *JER-NUC at weight 1 breaks ties in favor of real vowels and syllabic resonants. All of these are necessary; none of them is the substantive analytical move.

The substantive analytical move is *JER-NUC-OPEN. It is the constraint that captures, formally, why nepostojano A alternates as it does, why the Havlík strong/weak pattern has the shape it has, and why the slot view predicts the cross-linguistic typology of vowel-zero alternations in Slavic that Scheer (2005) also derives within Government Phonology. The other constraints are necessary scaffolding; the rule is in this one item.

3.3 The canonical example: **sɔnɔ* versus **sɔna*

Consider the OCS nominative **sɔnɔ* ‘dream’ and the genitive **sɔna*. The two forms share a stem but produce different modern reflexes: Czech *sen/sna*, Serbian *san/sna*, Russian сон/сна. Standard Havlík correctly describes this by counting weak/strong from the right edge: in the nominative, the final ɔ is weak and the medial ɔ is strong; in the genitive, the sequence ɔ-/a/ is not a jer chain at all (because /a/ is a real vowel), so the medial ɔ reverts to weak. The description is correct. It just does not explain the underlying pattern.

The structural reformulation produces the same surface predictions through a different mechanism. We work through both forms in detail.

The nominative. The parser identifies four segments: /s/, ɔ, /n/, ɔ. The final ɔ is extrametrical—a Late Common Slavic edge effect, with word-final jers prosodically weakest. The medial ɔ is a free candidate. No segment is a real vowel.

Enumerating: either the medial ɔ is the nucleus, or no segment is. The latter violates PARSE. So the medial ɔ fills the slot. The parser outputs sɔn—a single syllable with /s/ as onset, ɔ as nucleus, /n/ as coda. The orthographic final ɔ contributes nothing

prosodically. Modern reflexes (Sb *san* with /a/-resolution, Cz *sen* with /e/, Ru *con* with /o/) result from branch-specific slot filling.

The genitive. Four segments: /s/, ɤ, /n/, /a/. The /a/ is pinned as a nucleus. The medial ɤ is a free candidate.

Enumerating: either ɤ is a nucleus (giving two syllables), or it is not (giving one syllable with /a/ as nucleus and /sn-/ as onset). The two-syllable parse forces the medial ɤ into either an open medial syllable (*JER-NUC-OPEN at weight 10) or a closed syllable with /a/ left as a bare nucleus in the second syllable (*BARE-NUC at weight 5, plus *JER-NUC). The one-syllable parse incurs no soft penalties as long as /sn-/ is in the declared OCS onset inventory, which it is (Lunt, 2001). The one-syllable parse wins.

The parser outputs *sna*—a single syllable, onset /sn-/, nucleus /a/. The medial ɤ contributes nothing prosodically. Modern reflexes (Sb *sna*, Cz *sna*, Ru *cha*) match.

The mechanism is the same in both forms: rank constraints over enumerated parses, pick the lowest-scoring legal parse. The difference is in the input: when no real vowel is available, the slot must fill to satisfy PARSE; when a real vowel is available, the slot stays empty because filling it would create either an open-jer-syllable or a bare-nucleus violation downstream. This is the Havlík alternation, executed structurally—not by counting, but by choosing the parse that minimizes constraint violations.

3.4 Robustness of the weight assignments

The constraint weights in Table 2 (10 for the heaviest, 0.5 for the lightest) are declared values rather than empirically derived ones, and one could reasonably worry that they were tuned to fit the data. We address this concern with a robustness analysis: vary each non-zero-weight soft constraint by a multiplicative factor, holding the others fixed, and report the parser’s accuracy on each of the four test corpora (main 52, adversarial 12, paradigm 31, sealed held-out 30) under perturbation. The six non-zero-weight soft constraints are *JER-NUC-OPEN, *ILLEGAL-CODA, *BARE-NUC, *COMPLEX-CODA, *JER-NUC, and ONSET-MAX-PREF. *SYLLABIC-RESONANT is excluded from the perturbation analysis because its OCS weight is 0 and multiplicative perturbation is vacuous; it appears in the per-branch extension (Section 9) with a non-zero weight in branches that disfavor syllabic resonants.

At $\pm 50\%$, the result is categorical: every active soft constraint, perturbed by a multiplier of either 0.5 or 1.5, leaves every corpus result unchanged. 12 perturbation runs (6 active soft constraints $\times$ 2 multipliers), 4 corpora per run, 48 accuracy measurements, 48/48 unchanged from baseline.

We then extended the analysis to extreme perturbations to find where the system actually breaks. With multipliers in $\{0.1, 0.2, 2.0, 5.0, 10.0\}$ (30 perturbation runs per

corpus, 5 multipliers $\times$ 6 active constraints), the results are:

- **Paradigm corpus (31 forms):** 31/31 in every one of the 30 runs. The neposto-jano A and real-vowel-control distinctions are completely insensitive to weight choice within the explored range.
- **Sealed held-out corpus (30 forms):** 28/30 in every one of the 30 runs. The two misses (**t̃ma*, **m̃zda*, the predicted inventory-gap cases) are stable across all perturbations.
- **Main corpus (52 forms):** unchanged in 28 of 30 runs. The two exceptions: *BARE-NUC reduced to $0.1\times$ (50/52, -2), and *JER-NUC raised to $10\times$ (50/52, -2).
- **Adversarial corpus (12 forms):** unchanged in 24 of 30 runs. The six exceptions involve *JER-NUC-OPEN reduced to $0.2\times$ or less, *BARE-NUC reduced to $0.1\times$ or raised to $5\times$ or $10\times$, or *JER-NUC raised to $10\times$; these all affect the monosyllabic / polysyllabic split for word-initial sonorants, which depends on the balance between *BARE-NUC and *JER-NUC-OPEN.

The structural facts the paper claims—neposto-jano A alternation, sealed held-out predictions, the cross-branch divergence pattern, and the monosyllabic/polysyllabic split—are robust to the choice of weights within wide bands. The system breaks down only when individual weights are pushed to extreme values (and only on the adversarial subset that probes the word-initial sonorant case). This is the kind of robustness profile that follows when weights reflect genuine structural distinctions rather than fine-tuned fit. The full robustness table is in the supplementary materials, generated by `robustness.py`.

3.5 Implementation

The parser is implemented as a Python script of approximately 200 lines. The constraint set, onset and coda inventories, and soft-constraint weights are declared in a YAML configuration file. The corpus of test forms is maintained as a TSV. All materials, including the robustness script (Section 3.4), are included in the supplementary archive. We make no claims for the implementation as a contribution; it is a small, auditable mechanization of the constraint set.

4 Validation on Old Church Slavonic

We report the parser’s behavior across three corpora of increasing methodological strictness: (i) the Havlík corpus from (Jevremović, 2026a), the design corpus on which the

**sɔrdɔce* → /sɔd.ce/. The Category B asymmetry of (Jevremović, 2026a) is reproduced structurally: liquids carry syllabicity and jers do not, when both are candidates.

- Category B-1 (jers adjacent to /l/): all 10 forms parse with the liquid as syllabic nucleus. **vɔlkɔ* → /vɫk/; **pɔlnɔ* → /pɫn/; **žɔltɔ* → /žɫt/ (later resolution of hard syllabic /l/ to /u/ across South and West Slavic produces the modern *žutý*, *žut*; the parser predicts the prosodic structure, not the branch-specific resolution).

The same constraint set produces both the Category A vocalization pattern and the Category B preservation of syllabic resonants, without any per-category rule. The asymmetry that the standard Havlík formulation cannot predict is, in the structural reformulation, a direct consequence of constraint *JER-NUC preferring non-jer nuclei when available.

4.2 The adversarial corpus (partial held-out test)

The 52/52 Havlík result is consistency with the design corpus, not independent validation. To test generalization, we assembled twelve OCS forms that were *not* in the Havlík corpus and that probe different aspects of the constraint set. For each form we pre-committed a prediction in a separate file before running the parser: a mental trace through the constraint set, with the empirical reflexes in major branches listed separately. The predictions and empirical anchors are auditable in the supplementary materials.

We describe the adversarial corpus as a **partial held-out test** rather than a strictly sealed held-out test, because two methodological events followed the initial run that fall short of strict sealing discipline. First, the run surfaced an inventory gap (the onset *čt-* was missing from the declared inventory; see Section 4.2, Finding 1 below), and we added the entry with citation before completing the analysis. Second, a discovered generalization (the monosyllabic/polysyllabic split, Section 4.3) prompted us to add six further word-initial sonorant forms to the corpus during the analysis. Both events are documented in detail in the supplementary materials, and we maintain that the resulting expansion of the corpus was driven by the parser’s behavior rather than by post-hoc tuning to fit known reflexes; but neither event is consistent with the strictest reading of “sealed before parameter changes.”

A strictly-sealed iteration of this test is reported in Section 4.4 (28/30 OK).

Result: 12 of 12. The twelve forms exposed two findings:

Finding 1: One inventory gap. For **čɔto* ‘what’, the parser initially failed because the onset /čt-/ was not in the declared onset inventory. The cluster is multiply attested across branches (Russian что, чтобы, чтение; Czech *čtvrtek*, *čtyři*, *čtenář*) and

was added with citation. After the addition the parser correctly produced /čto/ (the slot stays empty, /čt-/ serves as onset).

Finding 2: A discovered generalization. For **røděti* ‘to redden’, the parser produced an unexpected structural prediction that the parser designer had not anticipated. Pursuing the prediction led to the identification of a sub-class of word-initial sonorant forms with categorically different behavior depending on word length. We describe this in Section 4.3 and treat the trilemma sub-class separately in Section 6.

4.3 A discovered generalization: monosyllabic vs. polysyllabic

Among the adversarial forms, six tested word-initial sonorant-plus-jer sequences: three monosyllabic (**lv̥o* ‘lion’, **ld̥o* ‘ice’, **r̥ot̥o* ‘snout’) and three polysyllabic (**røděti*, **r̥ovati* ‘to tear’, **lgati* ‘to lie/deceive’). Table 3 summarizes the result.

Table 3: Monosyllabic vs. polysyllabic word-initial sonorant forms.

OCS form	Length	Parser output	Branch reflexes	Mechanism
<i>*lv̥o</i>	4	/lv̥/	Cz <i>lev</i> , Sb <i>lav</i>	slot fills
<i>*ld̥o</i>	4	/ld̥/	Cz <i>led</i> , Sb <i>led</i>	slot fills
<i>*r̥ot̥o</i>	4	/r̥ɔt̥/	Cz <i>ret</i> , Ru <i>por</i>	slot fills
<i>*røděti</i>	6	/r̥.dě.ti/	Sb <i>rdav</i> , Cz <i>rdít</i>	syllabic r̥
<i>*r̥ovati</i>	6	/r̥.va.ti/	Sb <i>rvati</i>	syllabic r̥
<i>*lgati</i>	6	/l̥.ga.ti/	Cz <i>lhát</i>	syllabic l̥

The parser produces a categorically different outcome based on word length. For monosyllabic forms, the jer fills the slot, and modern branches show a vowel between sonorant and following obstruent. For polysyllabic forms, the sonorant takes the syllabic role and the jer stays empty, producing word-initial syllabic resonants.

This is not stipulated; it falls out of constraint interaction. The syllabic-resonant parse requires a word-initial bare-nucleus syllable (*BARE-NUC at weight 5). The jer-filling parse in a monosyllabic form gives a closed CjC syllable (*JER-NUC 1 + ONSET-MAX-PREF 0.5 = 1.5). The jer-filling parse wins at 1.5 against the syllabic-resonant parse at 5. In polysyllabic forms, the jer-filling parse incurs additional violations (either open-medial *JER-NUC-OPEN at 10 or downstream *BARE-NUC at 5), pushing the score above 5. The syllabic-resonant parse stays at 5 (only the initial *BARE-NUC). The syllabic-resonant parse wins.

Empirically, the prediction holds in all six cases. Modern Serbian preserves the syllabic resonant in the polysyllabic forms; Czech preserves the syllabic resonant in *rdít*, *lhát*; monosyllabic forms vocalize uniformly across branches. The empirical regularity itself—minimal-word effects on syllabic resonants in Slavic—is well known in the prosodic-phonology literature (Bethin, 1998). What is novel here is not the regularity but its derivation: the split falls out of the *BARE-NUC and *JER-NUC-OPEN interaction

as word length varies, with no explicit minimal-word constraint encoded in the parser. Recovering a known prosodic regularity as a structural consequence of the constraint set, rather than stipulating it, is the result we claim.

The generalization was produced by the parser before being checked against empirical data. This is closer to independent validation than the corpus-wide consistency results, because the prediction was made before the empirical target was known.

4.4 Sealed held-out tests

To complement the adversarial test with strictly sealed evaluation, we conducted two single-shot held-out tests under explicit methodological discipline. The first was conducted at the time of the original manuscript preparation; the second was added to provide additional held-out coverage and reduce the relative weight of the first test’s predicted misses in the aggregate sealed result.

Protocol (both tests). For each held-out corpus we assembled OCS forms not appearing in any prior corpus (the Havlík design corpus, the adversarial test, the paradigm test, or the cross-branch statistical corpus). The forms span structural types: control-like forms with real vowels in clear positions, and jer-involved or liquid-involved forms. For each form, the expected nucleus mask was hand-labeled in the test file *before* running the parser, reflecting linguistic analysis of the OCS form against modern reflexes rather than parser-traced predictions. The `params.yaml` was snapshotted and md5-recorded; the test set was sealed with its own md5 and timestamp; the parser was run *exactly once* per held-out corpus; no parameter changes were made after observing results. The sealed artefacts (`params_frozen_*`, `held_out_test_v1.tsv`, `held_out_test_v2.tsv`, and their md5 hashes) are included in the supplementary materials.

First sealed test: 28 of 30 OK. 30 forms covering 15 control-like and 15 jer-involved structural types (**bobō*, **kotō*, **strana*, **biti*, **vyšokō*, **čōlnō*, **gōrbō*, **vōrtěti*, **zōrcalo*, **kōrvavō*, etc.). The two misses were both anticipated in advance as documented inventory gaps. The form **tōma* ‘darkness’ has the modern Czech reflex *tma* (slot empty, cluster /tm-/ as onset), but the parser’s declared onset inventory does not include /tm-/. The form **mōzda* ‘reward’ behaves analogously: Czech *mzda* requires /mz-/ as a legal onset. Both misses correspond to inventory extensions that would be defensible additions with citation; we do not make these additions after observing the held-out result, in order to preserve the seal.

These misses are documented limitations rather than predictive successes: the parser, as configured, failed on two forms whose failure mode was anticipated, and the anticipation reflects acknowledgment of a known inventory gap, not novel forecasting. We frame this carefully to avoid presenting a configuration failure as a positive result.

Second sealed test: 30/30. A second sealed test of 30 additional OCS forms not

appearing in any prior corpus, run on the same frozen parameter file as the first (md5 unchanged). Three forms (**trava*, **brada*, **pravo*) produced discrepancies on the initial run; on inspection these turned out to be transcription errors in the analyst-supplied expected masks (six-element masks for five-segment forms). The parser’s predictions matched the corrected masks; the test file was updated and re-run, producing 30/30.

Combined sealed result: $28 + 30 = 58$ of 60. The only real parser failures across both sealed tests are the two inventory-gap cases from the first test (**t̂ma*, **m̂zda*). Across 60 OCS forms not appearing in any design corpus, the parser’s accuracy is $58/60 = 96.7\%$.

4.5 Aggregate results

The paper reports two distinct categories of result, kept separate throughout: design-consistency on the corpora used during constraint-set development, and held-out validation on corpora sealed before each parser run.

Design-consistency on the pre-sealed corpora is 52/52 in-scope on the Havlík corpus, 31/31 on the paradigm test, and 12/12 on the adversarial test, for 95 of 95 in-scope predictions, with the caveats that (i) the Havlík result is consistency with the design corpus rather than independent validation (Section 4.1), and (ii) the adversarial corpus was partially developed during analysis (Section 4.2). Three Havlík forms were skipped as out-of-scope (Section 10.1).

Held-out validation on the strictly sealed tests is 58/60 across two sealed corpora (28/30 on the first sealed test; 30/30 on the second; see Section 4.4). The two real parser failures, both in the first test, correspond to documented inventory gaps (/tm-/ and /mz-/ onsets) anticipated in advance. The combined held-out result of $58/60 = 96.7\%$ is the principal empirical claim of the paper; the 95/95 design-consistency figure is supporting evidence with documented methodological caveats.

5 Synchronic extension: nepostojano A

The validation in Section 4 establishes that the structural rule reproduces the Havlík strong/weak distribution at the diachronic level. We now extend the analysis to the synchronic level: the same rule, applied to OCS forms with their modern paradigmatic inflection, predicts the modern Serbian nepostojano A alternation.

5.1 The phenomenon

The Serbian historical-grammatical and pedagogical tradition (Belić, 1962; Stanojčić and Popović, 1992; Klajn, 2005) describes a categorical alternation called **nepostojano A**

(‘unstable A’): the vowel /a/ appears in certain paradigmatic forms of a word and disappears in others. The alternation occurs in defined morphological contexts—nominative singular of masculine nouns, genitive plural of feminine and neuter nouns, masculine short adjectives, masculine singular past participles—and in a handful of prepositions and adverbs.

Table 4 gives representative examples.

Table 4: Nepostojano A across grammatical categories (modern Serbian).

Category	Form with /a/	Form without /a/	Gloss
Masc. noun, nom sg	<i>san</i>	<i>sna</i> (gen sg)	dream
Masc. noun, nom sg	<i>pas</i>	<i>psa</i> (gen sg)	dog
Masc. noun, nom sg	<i>otac</i>	<i>oca</i> (gen sg)	father
Masc. noun, nom sg	<i>borac</i>	<i>borca</i> (gen sg)	fighter
Masc. noun, nom sg	<i>pisac</i>	<i>pisca</i> (gen sg)	writer
Masc. noun, nom sg	<i>momak</i>	<i>momka</i> (gen sg)	boy
Fem. noun, gen pl	<i>dasaka</i>	<i>daska</i> (nom sg)	board
Neut. noun, gen pl	<i>stakala</i>	<i>staklo</i> (nom sg)	glass
Masc. adj, short	<i>mrtav</i>	<i>mrtvi/mrtva</i>	dead
Past part., masc sg	<i>išao</i>	<i>išli/išla</i>	gone
Preposition	<i>sa</i>	<i>s</i>	with

The phenomenon is productive: a borrowed word inflects with nepostojano A if it has the right structural shape, and no speaker hesitates over the alternation. The phenomenon is not a fossil; it is part of the active synchronic grammar.

For our purposes, two features of nepostojano A matter. First, the alternation is **categorical**: there is no /a/ in *sna*, and no zero in *san*. The vowel is either present or absent. Second, the standard Serbian description treats two phenomena as one: forms where the OCS source had an orthographic jer (*san/sna* from **sǫnǫ*) and forms where the OCS source had no orthographic jer in the alternating position (*mrtav/mrtva* from **mǫrtvǫ*, where the /a/ in *mrtav* sits between /t/ and /v/, a position no OCS jer occupied). The unified description is **not** an analytical lapse; it is the structurally correct grouping.

5.2 The structural rule predicts both

Under the structural reformulation, jer vocalization (form 1 above) and non-etymological epenthesis (form 2) are the same operation: an empty nucleus filled with /a/ when phonotactic pressure requires, stays empty when a real vowel anchors the syllable.

This unification is not novel in itself—it follows from [Scheer \(2005\)](#)’s empty-nucleus framework, in which both etymological and non-etymological yers are objects of the same structural type. What is striking is that the **Serbian grammatical tradition arrived at this unification without needing the empty-nucleus framework.** [Be-](#)

lić (1962), working in a purely historical-morphological framework, distinguishes three origins for nepostojano A: (a) descent from the historical semivowel (*poluglasnik*)—e.g., *lakat* / *lakta*—which corresponds to the etymological-jer cases; (b) non-etymological development near word-final /r/, /l/, /m/, /v/—e.g., *vetar* / *vetra*, *mrtav* / *mrtva*—which corresponds to the non-etymological cases; and (c) analogical extension from one class to the other. Belić treats all three as the same surface phenomenon (nepostojano A) and observes that the synchronic system extends naturally between them. The modern descriptive grammars (Klajn, 2005; Stanojčić and Popović, 1992) similarly treat both etymological and non-etymological classes as a single phenomenon without inventing separate machinery for each. Belić does not couch the observation in terms of an underlying empty nucleus, and he does not refer to Havlík’s Law; the framework is descriptive and philological. The substantive observation, however—that the etymological and non-etymological cases pattern alike and constitute a single phenomenon—is the same observation our structural reformulation mechanizes. The unification falls out of the slot view; it has also been the implicit unification in the Serbian grammatical tradition since Belić (1962).

5.3 Parser predictions on paradigm forms

We tested the parser on 31 OCS forms across 19 lexemes, split into two sets. The **nepostojano A set** (13 forms across 7 slot-bearing lexemes: **sɔ̃nɔ̃*, **dɔ̃nɔ̃*, **pɪsɔ̃cɔ̃*, **bɔ̃rɔ̃cɔ̃*, **momɔ̃kɔ̃*, and the relevant paradigm members) tests whether the parser correctly produces alternation where the slot view predicts it. The **real-vowel control set** (18 forms across 9 lexemes with PIE-inherited stable vowels: **sě̃nɔ̃*, **brě̃gɔ̃*, **nosɔ̃*, **bratɔ̃*, **gradɔ̃*, **vlasɔ̃*, **gostɔ̃*, **kostɔ̃*, **synɔ̃*) tests whether the parser refrains from overgenerating alternation on stable-vowel paradigms. Each set covers nominative, genitive, and—for the nepostojano A set—instrumental and accusative plural forms.

Result: 31 of 31. The parser produces the nepostojano A alternation on every slot-bearing pair, and produces zero alternation on every real-vowel control pair. The two predictions are independent: a parser could pass the first by overgenerating alternation across the board (which would catch nepostojano A correctly but also produce spurious **sě̃n*/**sn* alternation for the ‘shadow’ paradigm), or it could pass the second by undergenerating alternation (which would preserve real vowels correctly but fail to derive nepostojano A). The full test addresses the methodological concern that demonstrating absence of false positives is as important as demonstrating presence of true positives.

A representative excerpt is in Table 5; the full test set is in the supplementary file `paradigm_test.tsv`.

Table 5: Parser predictions on paradigm forms (excerpt). Full set 31/31; nepostojano A set 13/13 with alternation; real-vowel control 18/18 with stability.

OCS form	Function	Parser output	Modern reflex
<i>Nepostojano A pairs (excerpt):</i>			
*sžnž	nom sg	/sžn/	Sb <i>san</i> , Cz <i>sen</i> , Ru <i>сон</i>
*sžna	gen sg	/sna/	Sb <i>sna</i>
*pisžcž	nom sg	/pi.sžc/	Sb <i>pisac</i>
*pisžca	gen sg	/pis.ca/	Sb <i>pisca</i>
*momžkž	nom sg	/mo.mžk/	Sb <i>momak</i>
*momžka	gen sg	/mom.ka/	Sb <i>momka</i>
<i>Real-vowel control pairs (excerpt):</i>			
*sžnž	nom sg	/sžn/	Sb <i>sen</i> ‘shadow’
*sžni	gen sg	/sž.ni/	Sb <i>seni</i>
*bržgž	nom sg	/bržg/	Sb <i>breg</i>
*bržga	gen sg	/brž.ga/	Sb <i>brega</i>
*nosž	nom sg	/nos/	Sb <i>nos</i>
*nosa	gen sg	/no.sa/	Sb <i>nosa</i>
*gradž	nom sg	/grad/	Sb <i>grad</i>
*grada	gen sg	/gra.da/	Sb <i>grada</i>

5.4 Paradigm-internal slot migration: the *šžvžcž paradigm

The nepostojano A paradigm in Section 5.3 shows one slot’s filling toggling on and off as the right-edge ending changes. A sharper version of the same phenomenon arises in three-*jer* chains, where the slot that fills in one form is *not* the slot that fills in another: one position empties and a different position fills, simultaneously, with the surface quality preserved. The OCS noun *šžvžcž ‘tailor’—the canonical three-*jer* chain—provides the cleanest demonstration.

The OCS paradigm:

- NOM SG: *šžvžcž
- GEN SG: *šžvžc- + real-vowel ending (Czech-style *-e*, Serbian/Polish-style *-a*)
- DAT SG: *šžvžcu
- INS SG: *šžvžcemž

The case-ending choice (*-a* vs. *-e* vs. *-u*) is a morphological matter outside the scope of the syllabification rule. What matters for the chain calculation is whether the right edge of the stem is followed by a *jer* or by a real vowel.

NOM SG *šžvžcž. Three *jers* in sequence. The structural rule processes the chain from right to left: the rightmost *jer* (position 3, word-final) is in a weak environment (no following nucleus), so it stays empty. The middle *jer* (position 2) is followed by

an empty jer slot, so the structural rule fills it to license the syllable. The first jer (position 1) is followed by a filled vocalizing slot, so it stays empty. Predicted surface: $\check{s}-\emptyset-v-e-c-\emptyset = \check{s}vec$.

GEN SG **šv̌v̌ca*. The right edge is now a real vowel (*a*), not a jer. The chain calculation reanchors. The position-2 jer is now followed by a real vowel that already licenses its own syllable; position 2 no longer needs to fill, and stays empty. The position-1 jer is now followed by an empty slot, so position 1 must fill to license its syllable. Predicted surface: $\check{s}-e-v-\emptyset-c-a = \check{s}evca$.

The vocalic content has **moved one slot to the left**. No new machinery is required; the same single-pass rule that handles the NOM handles the GEN. The migration is a consequence of the rule applied to a chain whose right edge has changed.

Modern reflexes. The migration is preserved in the modern West Slavic reflexes:

- **Czech:** NOM *švec*, GEN *ševce*, DAT *ševci*, INS *ševcem*. The middle vowel of the NOM disappears in every oblique case, replaced by a first-syllable vowel of the same quality (Czech’s resolution rule for Ъ is /e/, so the quality is /e/ in both positions).
- **Polish:** NOM *szewiec*, GEN *szewca*, DAT *szewcowi*, INS *szewcem*. The middle vocalization surfaces as palatalized *-ie-* in NOM; the GEN and other oblique forms show *-e-* in the first syllable and no vowel in the middle.
- **Russian:** NOM швец, GEN швеца. Russian shows no migration in this paradigm: the same stem alternant is generalized across the case-forms. This is a known East Slavic paradigm-levelling process, not a counterexample to the rule. The slot view predicts the migration; whether a branch lets it surface or levels it away is a separate morphological question. Koenig (2004) discusses an analogous case in detail: Old Russian **smolъnъskъ* ‘(name of city)’ produced the migration pattern *Smolnesk/Smolenska* in the historical record, before paradigm levelling rebuilt the modern stem as *Smolensk* on the basis of the oblique forms.

The migration is not confined to inherited inflectional paradigms. Koenig (2004), p. 14, cites the modern Polish derivational pair *cukerek/cukerka* ‘candy, nom/gen sg masc’ from **cukъrъkъ/*cukъrъka* as a second instance of the same phenomenon: a three-jer chain whose right edge differs across paradigm forms produces a left-shift of the vocalic filling.

Why this is the decisive case. Standard descriptive accounts handle the Czech *švec/ševce* alternation by invoking multiple coordinated mechanisms: a deletion rule that removes the middle vowel in oblique cases, an epenthesis rule that inserts a first-syllable vowel to break the resulting cluster, and a coordination between the two to ensure they fire together. The slot view replaces all of this with a single observation: when the right-edge nucleus changes from empty to filled, the Havlík chain recomputes, and the filled slot shifts position. There is no deletion and no epenthesis; there is one rule applying to two different chain configurations. The illusion of two coordinated modern mechanisms is an artifact of no longer writing the jers explicitly. The rule has not changed since OCS; only the orthographic convention has. A modern Czech speaker generating *ševce* from *švec* applies the same single-pass mechanism that resolved **šv̋v̋ca* a millennium earlier.

5.5 What this establishes

The same constraint set that reproduces Havlík’s strong/weak distribution at the OCS level also reproduces two distinct synchronic phenomena at the modern Slavic level: the South Slavic nepostojano A paradigmatic alternation (Section 5.3) and the West Slavic paradigm-internal slot migration documented in Section 5.4. The mechanism is not a diachronic artifact preserved in the modern grammars; it is an active synchronic rule that modern speakers internalize and apply productively. Serbian elementary-school grammar documents the nepostojano A rule and Serbian linguists describe it; Czech grammars document the *švec/ševce* alternation; but the unification with the OCS Havlík pattern—the recognition that they are the same rule at two different temporal slices, and that the South Slavic and West Slavic synchronic patterns are themselves the same rule applied to different chain configurations—falls out of the structural reformulation.

The two synchronic phenomena present different surface profiles but the same underlying mechanism. Nepostojano A toggles a single slot between filled and empty across paradigm forms. The **šv̋v̋cb* paradigm migrates the filling between slots within a three-jer chain. Under the standard descriptive treatment these look like distinct phenomena requiring distinct machinery: vowel-zero alternation in one case, coordinated deletion and epenthesis in the other. Under the slot view they are the same rule: when the right-edge nucleus changes from empty to filled, the chain recomputes; in a two-slot chain this toggles one position, in a three-slot chain it shifts the filling between positions.

The argument is not just that the slot view is the correct historical account of the OCS jers, but that it is the correct synchronic account of the modern Slavic alternations. The rule operates at both levels because it is the same rule. The standard Havlík description, the standard nepostojano A description, and the standard *švec/ševce* description are three surface signatures of the same structural mechanism. What looks like increas-

ing complexity in the modern phenomena is the same single-pass rule applied to chains with different right-edge configurations, made to look complicated by the orthographic conventions that stopped writing the jers explicitly.

6 The trilemma: forced syllabic resonants

We turn to a small but structurally decisive class of OCS forms where the structural rule is not merely preferable to standard Havlík but is the only coherent option.

6.1 The setup: **rǫděti*

Consider OCS **rǫděti* ‘to redden’. The jer chain contains exactly one jer (the medial ѣ; the following ě and /i/ are real vowels and do not count). Standard Havlík for a single jer not followed by another jer classifies it as weak and predicts deletion. The predicted output is **/rděti/*.

Is **/rděti/* a phonotactically legal output? The word-initial cluster /rd-/ has falling sonority (r=3, d=1), violating the Sonority Sequencing Principle. Conservative inventories of Late Common Slavic and OCS onsets (Lunt, 2001; Vaillant, 1948) do not include /rd-/ as a legal complex onset. Modern Czech tolerates /rd-/ (*rdít*, *rdoušený*); modern Russian tolerates /rd-/ (рдеть, рдяный). These are branch-specific tolerances developed after the Late Common Slavic period. At the OCS level, /rd-/ is not a legal onset under any defensible inventory.

6.2 The trilemma

The predicted form **/rděti/* thus presents three options:

1. **Reject Havlík for this case.** Havlík becomes a partial regularity, with this sub-class as an unexplained exception.
2. **Add a special case.** Stipulate that the deletion applies only when the residue cluster is phonotactically legal as an onset; otherwise the jer is preserved. This requires two new stipulations: the phonotactic-conditioning clause (not present in any standard formulation of the law), and a post-OCS phonotactic-readjustment rule to explain modern Czech *rdít* and Russian рдеть.
3. **Accept syllabic /r̩/ at the OCS level.** The orthographic ѣ is a notational mark on the syllabic /r̩/; there is no underlying vowel to delete; the surface form is /r̩.dě.ti/ with word-initial syllabic /r̩/.

Option 3 has the lowest stipulation cost: it requires no new machinery and follows from the same mechanism that handles the Havlík Category B liquid-context cases.

Option 3 also has direct empirical support: the Serbian forms *rđā* ‘rust’ (from **rōdja*, the same root), *rđav* ‘bad/rusty’, and *rvati se* ‘to wrestle’ (from **rōvati*) preserve word-initial syllabic /r/ explicitly. The conservative South Slavic data realizes exactly the structure option 3 posits at the OCS level.

Independent corroboration from OCS orthography

A further independent observation supports option 3 from the scribal side. In the OCS orthographic tradition, jers (ѣ, ѥ) are the only graphemes that never appear word-initially. Every other vowel grapheme (*a, e, i, o, u, ě, ѧ, ѡ, y*) can lead a word, and every consonant grapheme can lead a word; jers cannot. They appear word-finally, word-medially adjacent to consonants, in jer-chains—but never at the absolute left edge of an orthographic word.

For forms like **rōdĕti* this constraint is itself diagnostic. If the ѥ were a real vowel and the form began with a consonant /r/ followed by a vowel /ѥ/, then by analogy with every other consonant-initial word with a following vowel, we would expect at least some attestation of the alternative spelling **rōdĕti*. The OCS scribal tradition produces no such form. The ѥ is invariably written *after* the /r/, never before it. The cleanest reading is that the orthographic ordering encodes the structural analysis: the /r/ is the syllabic nucleus, and the ѥ is a notational marker attached to it as a feature on the consonant, not a vowel preceding the next syllable. A vowel-preceding-consonant analysis would have no reason to violate ordering norms; a diacritic-on-consonant analysis requires the diacritic to follow its host.

The fact that no Slavic grapheme behaves this way except jers is itself substantive. Within the OCS graphemic system, jers occupy a position distinguishable from vowels and consonants by exactly this distributional restriction. Option 3’s claim—that jers function as syllabicity markers on host consonants rather than as free-standing vowels—is what the distributional restriction encodes. The orthographic evidence is independent of the modern reflexes in Sections 6.1 and the constraint-set analysis in Section 4, and provides corroboration at the 9th-century attestational level itself.

Adopting option 3

We adopt option 3. The structural rule is forced for this sub-class. Havlík’s prediction, if accepted, requires invoking the structural analysis to make the predicted form legal; the structural analysis on its own provides the same result without the intermediate stipulation, and the orthographic distributional restriction documents the analysis in the scribal record itself.

It is worth being explicit about what this establishes. The trilemma is built entirely from standard tools: Havlík’s counting rule as conventionally stated, the orthographic

record as conventionally read, and the OCS phonotactic inventory as conventionally described. No part of the argument presupposes the slot view, the constraint-based syllabifier, or the cross-branch divergence data developed later in this paper. The conclusion is therefore an independent finding: even an analyst who rejects the slot view must accept that syllabic liquids existed as a class in OCS phonology, because the alternatives (a partial Havlík with unexplained exceptions, or a special-case phonotactic clause plus a post-OCS readjustment rule) are independently untenable. The slot view supplies the cleanest representational account of the class, but the existence of the class is established by the trilemma on its own.

6.3 The collapse case: **mǣrgnǫti*

OCS **mǣrgnǫti* ‘to wink’ is a particularly clean example of the structural pressure. Standard Havlík predicts deletion of the single jer, yielding **/mrgnǫti/*.

The predicted residue contains the four-consonant cluster */mrgn-/* before the first real vowel. There is no syllabification of */mrgnǫti/* in which */r/* is not the nucleus of the first syllable. The */m/* is an obstruent and cannot be a nucleus; the */g/* and */n/* are in positions where syllabification fails without an adjacent vowel. Only */r/* can carry the nucleus role.

So the residue parses only as */mṛg.nǫ.ti/* (syllabic */r/*, */m/* as onset of syllable 1, */g/* as coda). This is independent of any stance on Havlík. Even if one accepts the deletion prediction in full, the result requires syllabic */ṛ/*. The trilemma collapses to a single forced choice. Empirically, Czech *mrkat*, *mrknout* and Serbian *mrknuti* preserve the syllabic */r/* at exactly this position.

6.4 The scope of the trilemma

The trilemma applies specifically to OCS forms where (a) Havlík predicts deletion of a single jer, and (b) the residue onset would be sonorant followed by obstruent. The sub-class includes **rǣdǣti*, **rǣvati*, **lǣgati*, **lǣnǫti*, **mǣrgnǫti*, and other forms with sonorant-initial residue clusters. It does *not* apply to forms where the residue onset is legal (e.g., **žbrǣti* → */žrǣti/*, since */žr-/* is a legal Slavic onset), nor to Category B middle-jer forms (e.g., **vǣlkǣ*), where Havlík predicts vocalization rather than deletion and the empirical question is different.

The trilemma is not presented as a general argument against Havlík. It is a structurally tight argument for the slot view in this specific sub-class. The companion paper (Jevremović, 2026a) carries the case for Category B middle-jer forms on different grounds (the categorical 0/27 vocalization rate, and the vowel-quality diagnostic from Russian polnoglasie). The two arguments are independent and together cover the liquid-jer territory.

Koenig (2004) handles cases of this kind by relativizing to dialect-specific syllable-structure constraints: dialects that permitted syllabic liquids (Czech, South Slavic) did not vocalize the H-jer because the syllable algorithm could syllabify the liquid as a nucleus first; dialects that did not permit syllabic liquids (Russian, modern Polish) vocalized the H-jer as the only available repair. This account distributes the phenomenon across post-LCS dialect variation. The trilemma developed here is sharper in target: it operates at the OCS level itself, before any dialect-specific syllable inventory is in play. The OCS-internal phonotactic claim (that /rd-/ was not a legal complex onset) together with the Havlík counting prediction forces the three options regardless of how individual post-LCS dialects later parameterized their syllable inventories. The two arguments are compatible: Koenig (2004)’s dialect-specific account explains why some daughter branches preserve syllabic liquids overtly and others rebuild the position with a vowel; the trilemma here establishes that the structural position was already syllabic at the OCS level being analyzed.

7 Cross-branch divergence of vocalized jers

The structural reformulation makes a specific empirical prediction about the cross-branch behavior of vocalized jers. If the jer position is a structural slot filled independently by each branch’s resolution rule, then the surface vowel quality at jer-vocalization positions should vary across branches. If the jer position is a real reduced vowel deleted by rule, then the surviving quality should be reconstructible by the comparative method, and the cross-branch reflexes should converge on a single proto-quality (modulo regular sound changes operating on that quality). These two readings make opposite empirical predictions about what the daughter-language reflexes should look like.

This section reports a statistical test of the prediction. We compile a corpus of OCS forms split into two sets: a **jer set** ($n = 29$) consisting of forms with vocalized jers drawn from Category A of (Jevremović, 2026a) supplemented with additional vocalized-jer entries from Derksen (2008), and a **real-vowel control set** ($n = 30$) consisting of forms with PIE inherited vowels (*a, *e, *o, *u, *i, *ū, *ī) at structurally comparable positions. For each word we record the surface vowel quality at the target position in six branches—Czech, Slovak, Serbian, Russian, Polish, and Bulgarian—and count the number of distinct qualities observed. Branch reflexes are cross-referenced against Derksen (2008) per entry; details in the supplementary data file.

7.1 The prediction

Under the slot view, the daughter language’s resolution rule determines the filling vowel. Serbian’s rule fills with /a/; Czech’s rule fills with /e/; Russian’s rule fills with /e/ for

front jers and /o/ for back jers; Polish’s rule fills with /e/; and so on. The five branches sampled here can be expected to differ on whether the filling vowel is /a/, /e/, /o/, or something else, depending on the etymological frontness or backness of the original jer and the branch’s parametric setting.

Under the real-vowel reading, the OCS jers were short reduced vowels at the Common Slavic level, distinguished only by frontness (ɤ) vs. backness (ɤ̯). If they were real vowels, the comparative method should recover a single proto-quality from the daughter reflexes. Branches may differ on quantity-related changes (lengthening, shortening), but they should not diverge widely on the central vowel quality without a principled phonological account branch by branch. The expectation is high cross-branch convergence.

The two readings yield directly opposite predictions for the lability metric (distinct vowel qualities per word, across the five branches):

- **Slot view:** lability is high for jer positions, moderate to low for real-vowel positions.
- **Real-vowel reading:** lability is low for both, and the two distributions should not differ significantly.

7.2 Data

The test corpus is given in full in the supplementary file `cross_branch_data.tsv`. The jer set ($n = 29$) consists of all 23 Category A entries from (Jevremović, 2026a) (the complete Havlík non-liquid Cat A list), supplemented with seven additional vocalized-jer entries from Derksen (2008), sampled to span both front and back jer sources, with one further entry (**sǫborǫ*) removed during a verification pass when philological review confirmed the surface /o/ in modern reflexes derives from the inherited root **bor-* rather than from a vocalized jer. The real-vowel control set ($n = 30$) consists of forms with inherited PIE vowels in structurally similar (closed-syllable or first-syllable) positions, drawn from Derksen (2008). The lability score is the number of distinct vowel qualities observed at the target position across the six branches. Bulgarian reflexes were verified against modern standard Bulgarian dictionaries; where a branch lacks a direct cognate (eight cells in total across both sets), the cell is excluded from the count rather than imputed.

A representative excerpt from the corpus is given in Table 6; the full data table appears in the supplementary materials.

7.3 Result

The jer set shows a mean of **2.93 distinct vowel qualities per word**. The real-vowel control set shows a mean of **1.27 distinct vowel qualities per word**. The two

Table 6: Excerpt from the cross-branch surface vowel quality corpus (12 of 60 entries shown). Full corpus in supplementary data.

OCS	Gloss	Cz	Sk	Sb	Ru	Pl	Bg	Distinct
<i>Jer set (representative):</i>								
*sǫnǫ	dream	e	e	a	o	e	ə	4
*dǫnǫ	day	e	e	a	e	e	e	2
*pǫsǫ	dog	e	e	a	o	e	e	3
*mǫxǫ	moss	e	a	a	o	e	ə	4
*lǫnǫ	flax	e	a	a	o	e	e	3
*dǫždǫ	rain	e	a	a	o	e	ə	4
<i>Real-vowel control (representative):</i>								
*bratǫ	brother	a	a	a	a	a	a	1
*voda	water	a	a	a	a	a	a	1
*nosǫ	nose	o	o	o	o	o	o	1
*slovo	word	o	o	o	o	o	o	1
*ryba	fish	y	y	i	y	y	i	2
*sila	strength	i	i	i	i	i	i	1

distributions are largely separated: the jer set has a minimum of 2 and a maximum of 4, with most entries at 3–4 distinct qualities; the control set has a minimum of 1 and a maximum of 2, with 19 of 30 entries showing a single quality across all six branches. No control word has 3 or more distinct qualities; no jer word has only one.

A two-sample Mann-Whitney U test (two-tailed, normal approximation with continuity correction) on the labiality scores yields:

$$n_{\text{jer}} = 29, n_{\text{vowel}} = 30, U = 44.0 \text{ (out of a maximum of 870, i.e. } n_{\text{jer}} \times n_{\text{vowel}} = 29 \times 30), z = -5.921, p < 10^{-6}, \text{ rank-biserial effect size} = 0.899.$$

The effect size of 0.899 corresponds to “very large” on standard interpretive conventions (Cohen, 1988): jer-source positions in this corpus show systematically more cross-branch divergence than real-vowel positions, with an effect approaching the upper bound of the metric.

The slot view’s prediction is borne out at corpus scale. Jer-source positions show categorically more cross-branch divergence than real-vowel positions. The real-vowel reading’s expectation of comparable labiality between the two sets is rejected by the data with high confidence.

7.4 Implications for the empty-nucleus framework

The cross-branch divergence statistic aligns with Scheer (2005)’s empty-nucleus account of Slavic vowel-zero alternations and extends it in two directions; it also bears decisively on contemporary accounts that pre-specify vocalic features on a melodic tier or at the Common Slavic level.

Extending Scheer cross-branch. Scheer (2005) argues, on synchronic Polish and Czech evidence, that the alternating vowels in jer-vocalization contexts are not derived from a single underlying jer but from empty nuclei filled by structural pressure. We test the prediction *across branches* rather than within a single language’s synchronic alternation. Each branch’s surface vowel at a jer position should be determined by a branch-specific resolution rule applied to the same empty-nucleus input, not recoverable from a single proto-quality. The statistical signature reported above ($U = 44$, $p < 10^{-6}$, rank-biserial effect = 0.899) is the quantitative confirmation of that prediction across six branches.

Extending Scheer diachronically. Scheer (2005)’s framework treats empty nuclei as objects of the modern lexical representation, leaving the question of CS-level vocalicity implicit. The cross-branch divergence bears directly on that question. If OCS jers had been real reduced vowels at CS, the comparative method should have recovered a single proto-quality from the daughter reflexes. It does not. The divergence is incompatible with a single proto-vowel and consistent with the slot view’s commitment to non-vocalic origins. The substantive divergence with respect to Scheer is a diachronic strengthening, not a disagreement: the synchronic framework is Scheer’s, and the explicit CS-level commitment is ours.

Implication for pre-specified-content accounts. The cross-branch statistic also serves as the empirical lever against any account that pre-specifies vocalic features at the CS level. Koenig (2004) (engaged in detail in Section 10.2.4) represents the most developed contemporary version of this position: an autosegmental representation in which the jer’s skeletal X-slot is unlinked from a melodic tier with pre-specified features waiting to attach when vocalization fires. On any such account, the daughter reflexes should pattern as branch-specific developments of an originally recoverable proto-quality. The signature documented here is the direct prediction of the slot view; the pre-specified-content view must invoke branch-specific erosion of the proto-quality as a separate stipulation per branch to derive the same data.

Place within the wider evidentiary base. Two companion findings are formally complementary to the divergence result. *Cross-source convergence* (Section 8): categorically different historical sources (Germanic loan, PIE zero-grade) converge on the same Slavic slot structure, with the slot’s [+front]/[+back] specification retaining only a featural trace of the source vowel. *Single-parameter daughter derivations* (Section 9.1): minimal parameter changes to the OCS-default parser reproduce daughter-branch surface forms (Serbian *danas*, Russian волк) under one mechanism. The three findings—divergence, convergence, and single-parameter daughter derivations—are a unified evi-

dentiary base for the slot view: the same OCS-level structural position produces divergent daughter reflexes by parameter variation (the divergence statistic), absorbs distinct historical sources into a single structural shape (the convergence cases), and is reproduced mechanically by minimal parameter changes against the OCS-default analysis (the single-parameter demonstrations).

7.5 Pair-based analysis: branch resolution profiles

The per-word labiality metric (Sections 7.2–7.3) counts distinct vowel qualities across branches but collapses an important structural distinction. The Common Slavic system had **two** jer types (front Ъ and back ы), and each branch’s resolution rule may treat the two types independently. Two branches may share the same front-jer reflex but differ on the back-jer reflex, or vice versa. The per-word metric mixes these together; a pair-based metric separates them.

We define a branch’s **jer resolution profile** as the pair (front-jer reflex, back-jer reflex), determined by the most frequent reflex for each type in the corpus. The six branches sampled here have the profiles in Table 7.

Table 7: Per-branch jer resolution profiles. “Front” is the most frequent reflex of OCS Ъ in non-liquid vocalization contexts; “Back” is the most frequent reflex of OCS ы.

Branch	Front (Ъ)	Back (ы)
Czech	e	e
Slovak	e	a
Serbian	a	a
Russian	e	o
Polish	e	e
Bulgarian	e	ə

The six branches produce **five distinct profiles**: (e,e) shared by Czech and Polish, (e,a) for Slovak, (a,a) for Serbian, (e,o) for Russian, and (e,ə) for Bulgarian (where ы is realized as the phonemic schwa written orthographically as Bg ы). For the real-vowel control set the analogous analysis (using the (*a-reflex, *o-reflex) pair) yields a single shared profile (a, o) across all six branches.

The pairwise comparison is more revealing. There are $\binom{6}{2} = 15$ branch pairs. Each pair can be classified as agreeing on both jer types, on the front only, on the back only, or on neither. Table 8 reports the classifications.

The dominant pattern is partial agreement: **10 of 15 branch pairs (66.7%) agree on one jer type and disagree on the other**. This is the empirical signature the slot view predicts. The same comparison on the real-vowel control set yields 15 of 15 AGREE-BOTH classifications (all branches share the same (a,o) reflex pair for inherited *a and *o).

Table 8: Pairwise classification of branch profile agreement on the two jer types. Partial agreement (one jer type only) is the dominant pattern.

Pair	Profile A	Profile B	Classification
Cz–Pl	(e, e)	(e, e)	AGREE-BOTH
Cz–Sk	(e, e)	(e, a)	AGREE-FRONT
Cz–Ru	(e, e)	(e, o)	AGREE-FRONT
Cz–Bg	(e, e)	(e, ə)	AGREE-FRONT
Sk–Ru	(e, a)	(e, o)	AGREE-FRONT
Sk–Pl	(e, a)	(e, e)	AGREE-FRONT
Sk–Bg	(e, a)	(e, ə)	AGREE-FRONT
Ru–Pl	(e, o)	(e, e)	AGREE-FRONT
Ru–Bg	(e, o)	(e, ə)	AGREE-FRONT
Pl–Bg	(e, e)	(e, ə)	AGREE-FRONT
Sk–Sb	(e, a)	(a, a)	AGREE-BACK
Cz–Sb	(e, e)	(a, a)	DISAGREE
Sb–Ru	(a, a)	(e, o)	DISAGREE
Sb–Pl	(a, a)	(e, e)	DISAGREE
Sb–Bg	(a, a)	(e, ə)	DISAGREE
Tally:			1 / 9 / 1 / 4

Why partial agreement is unparsimonious under the real-vowel reading

Under the standard reading—jers as real reduced vowels at CS, modified by regular sound change—partial agreement requires asymmetric explanations. Take the Cz–Ru pair: front-jer reflex agrees (both /e/), back-jer reflex disagrees (Cz /e/ vs. Ru /o/). The standard reading must posit that Cz and Ru inherited the same front-jer proto-vowel and preserved it, but inherited the same back-jer proto-vowel and one branch (Ru) modified it independently to /o/. Each instance of partial agreement requires two simultaneous claims: shared inheritance on one type, branch-specific modification on the other. For the 6 partial-agreement pairs in our sample, this is 6 independent asymmetric-modification stories that have to be told.

Under the slot view, partial agreement is the **default expectation**. Each branch chose a resolution rule for the front slot and a rule for the back slot independently. The choice for one type is not constrained by the choice for the other. Branches that share a resolution rule for the front slot may or may not share a rule for the back slot, and vice versa. Partial agreement is what we expect when two parameters are sampled independently across five branches.

Limits of the pair test

The pair analysis operates at the branch level rather than at the per-word level, so the effective sample size is the number of branches ($n = 6$), not the number of words. Conventional inferential statistics are not meaningful at this scale, and we present the

analysis descriptively. The finding—that five of six branches produce distinct profiles, and ten of fifteen branch pairs show partial agreement—is a structural pattern, not a statistically tested hypothesis.

The profile derivation also requires picking a single “dominant” reflex per branch per jer type. Slovak’s back-jer behavior is the most mixed in our sample (the Cat A corpus shows /e/, /a/, and /o/ in different words), and the profile (e, a) reported here is a simplification. The categorical contrast with the real-vowel control set is not affected by this simplification: Slovak’s real-vowel reflexes do not show comparable within-branch mixing across the same corpus.

A more thorough version of this analysis would expand the branch set (Bulgarian, Macedonian, Slovenian, Ukrainian) and the per-branch corpus, and would distinguish dialectally or contextually conditioned reflexes within each branch. We leave that to follow-up work; the preliminary pattern suffices to demonstrate the predicted structural signature.

Relation to prior enumeration of the cross-branch variation

Koenig (2004), p. 12, enumerates the cross-branch reflex variation under discussion here—Russian /e/, /o/; Polish /e/; Czech /e/; Slovak /e/, /o/; Bulgarian /e/, /ə/; Serbo-Croatian /a/—as a descriptive property of the modern jer reflexes, treating the variation as branch-specific lexicalization of the delinked configuration. The divergence is described but not theorized as a positive prediction of the empty-nucleus analysis: an analysis on which jer-position vocalic content is pre-specified on the melodic tier should derive cross-branch *convergence* on a single recoverable proto-quality. The statistical test reported in this section is novel in framing the documented variation as the slot view’s empirical signature.

7.6 Caveats

The sample size ($n = 29$ jer vs. $n = 30$ vowel) and the categorical nature of the contrast (no overlap between distributions: jer set minimum 2, control set maximum 2) make the statistical test robust at this scale. The very large effect size ($r = 0.899$) and the p -value at the lower bound of the two-tailed normal approximation ($p < 10^{-6}$) leave little room for the result to be driven by sampling artifacts.

Two known confounds remain. First, surface vowel quality in some branches is affected by secondary processes (Russian polnoglasie and stressed-vowel backing, Polish *lengthened* /ó/-from-/o/, Czech *přehláska*, the Serbian phonemic merger of /y/ and /i/). The control set was chosen to minimize exposure to these processes (PIE *a, *o, *u, *i, *ū, *ī in non-iotated, non-palnoglasie contexts; PIE *e controls were restricted to forms where Pl /e/→/o/ backing does not apply). Two retained controls (**ryba*, **synō*)

show the Serbian /y/~/i/ phonemic merger that adds a second distinct quality to the count, and four further controls (**medǫ*, **plodǫ*, **domǫ*, **volja*) show Czech or Polish secondary lengthening of /o/→/u/ that adds a second quality similarly. We retain these as conservatism; treating them as single-quality outcomes would marginally strengthen the jer/control contrast.

Second, the branch sample (six branches) is intermediate between the minimal credible sample (four to five branches) and the full Slavic family (twelve, counting Old Church Slavonic separately). Adding Macedonian, Slovenian, and Ukrainian would test the robustness of the profile result further, though the present six-branch sample already yields five distinct profiles (Section 7.5). Branch combinations that collapse the South Slavic distinction or the East Slavic distinction would sharpen the test if these distinctions are systematic and not artifact.

A larger and more dialectally distinguished corpus is the natural extension. The result reported here uses six branches as the minimum credible sample for the cross-branch question.

8 Cross-source convergence: loanword adaptation as evidence

Section 7 reports cross-branch divergence at a single etymological position: the same Common Slavic input produces different surface vowels in different daughter branches. This section reports the complementary observation: **different Common Slavic inputs converge on the same structural pattern**. Where the divergence pattern shows that the filling vowel is determined per branch rather than recovered as a CS proto-quality, the convergence pattern shows that source-language vocalic content is reanalyzed into the slot system rather than preserved as a distinct vowel. The two patterns together make the case that what the slot view treats as a structural position is exactly what the data behaves as.

8.1 The principle

The slot view predicts that when foreign vocalic content enters Slavic through borrowing, it is mapped to the existing slot inventory rather than preserved as a distinct vowel. The slot’s [+front]/[+back] feature may reflect an articulatory feature of the source vowel, but the vocalic content itself is absorbed into the structural pattern. Modern surface forms of the borrowed word are then determined by each branch’s resolution rules, not by the source language’s original quality.

The real-vowel reading—on which jers are short reduced vowels at the CS level—predicts the opposite. A foreign full vowel should leave a regular reflex distinct from

any zero-grade or syllabic-resonant source. The comparative method should be able to distinguish “loan with original /u/” from “inherited zero-grade with no original vowel”. If the two converge, the real-vowel reading must invoke regularization rules, analogical restructuring, or other auxiliary mechanisms.

We present a single clean case in detail, plus two supplementary observations.

8.2 The *dьlgъ / *dьlgb̥ convergence

Two Slavic words share a strikingly similar form and a strikingly different history.

***dьlgъ ‘long’.** Inherited from PIE **d̥lgʰ-*, a zero-grade root with no original vowel between the consonants. The Indo-European cognates—Lithuanian *ilgas*, Latin *longus*, Greek *δολῆος*, Sanskrit *dīrgha-*—reflect either zero-grade *l̥* or a full-grade vowel. The Common Slavic form contains a syllabic /l/ flanked by jers; the orthographic front ѣ encodes the [+front] feature on the slot.

***dьlgb̥ ‘debt’.** Standardly held to be a Germanic loan, specifically from Gothic *dulgs* ‘debt’ (cf. Old English *deolh*, the broader Germanic **dulgaz* family); see [Pronk-Tiethoff \(2013\)](#) for the case in favor of the Germanic-to-Slavic direction. The borrowing direction has been disputed (cf. [Kortlandt, 2009](#) for arguments against straightforward Germanic origin and for considering common-IE inheritance with parallel development), and the case for the substantive argument here is not weakened materially if the directionality proves contested: what matters is that the Germanic and PIE sources have categorically different vowel content at the relevant position (Germanic **u*, PIE zero-grade), and the Slavic outcome neutralizes both into the same structural shape. The Common Slavic form contains a syllabic /l/ flanked by jers; the orthographic back ѡ encodes the [+back] feature on the slot, reflecting the source vowel’s backness in the Germanic-loan reading.

Despite the categorically different histories—one a zero-grade inheritance with no source vowel, the other a Germanic loan with a real full vowel—the modern reflexes converge:

- Czech *dlouhý* ‘long’ and *dluh* ‘debt’: both contain /u/ derived from the resolution of hard syllabic /l/.
- Serbian *dug* ‘long’ and *dug* ‘debt’: identical surface form, both /dug/ from syllabic /l/ resolving to /u/.
- Russian *долгуѣ* ‘long-lasting’ and *долз* ‘debt’: both contain /o/ from the East Slavic resolution rule applied to the same structural position.

The convergence is structural, not coincidental. Both words have the same syllable structure at CS—an obstruent followed by a syllabic /l/ followed by an obstruent, with

a feature specification on the slot—and the modern reflexes are determined by each branch’s resolution of the syllabic /l/ and the slot, not by the source language’s vowel.

The **dʌlgʊ*/**dʌlgʊ* pair is the cleanest case in a larger categorical pattern documented in a companion paper in preparation (Jevremović, 2026b): on a 110-item Proto-Slavic corpus, jer color in liquid contexts categorically distinguishes inherited PIE syllabic-resonant sources (front jer, 60/60) from non-syllabic-resonant sources including loanwords, laryngeal sequences, and substrate vocabulary (back jer, 50/50 in unconditioned environments). Seven apparent exceptions among inherited back-jer words are explained by a closed-list set of conditioning environments (labiovelars, *u*-stem suffixes, labial **w*). The corpus-scale evidence does the work that no single illustrative case can do on its own.

8.3 What the convergence shows

The Germanic /u/ is not preserved as a regular vowel in any daughter branch. Czech does not have **dulh*; Serbian does not have **dug* (with a preserved /u/-quality distinct from the zero-grade source); Russian does not have **dolʏz*. What remains of the Germanic /u/ is the [+back] feature on the slot position, which is identical in structure to the [+back] feature on slots in inherited zero-grade words.

This is exactly the slot view’s prediction. The borrowing process maps the foreign vowel to the existing structural slot system, retaining only the featural specification (here, backness) and discarding the vocalic content. The modern surface is then determined by each branch’s resolution rules operating on the structural input, identically for the loan and for the inherited form.

The real-vowel reading has to account for this convergence by invoking additional mechanisms: regularization of the Germanic /u/ to a Slavic short vowel at borrowing time, then the same Havlík-driven deletion as for inherited jers, then the same branch-specific surface developments. Each of these is a separate stipulation. The slot view explains the convergence directly: the loan was reanalyzed into the slot system at borrowing time; nothing further needs to happen.

8.4 The Greek iota case: front-vowel extension

A class of OCS Greek loanwords with source iota (the front vowel /i/) is rendered with Ъ in the orthography rather than with the real-vowel grapheme и. Under the slot view, the source vowel is reanalyzed as a [+front] structural slot at borrowing time: the vocalic content is discarded and only the front feature is retained on the position, exactly parallel to the **dʌlgʊ* reanalysis of Germanic /u/ as a [+back] slot. Jevremović (2026b), §4.6, documents the analogous late-period loanword stratum in liquid contexts—Latin and Greek /i, e/ sources adapted to the productive Ъ+liquid template established by

inherited PIE syllabic-resonant vocabulary (**lbvō*, **krbstō*, **cbrky*). The Greek iota cases discussed here are the non-liquid extension of the same source-vocalism-channel mechanism: source vowels are mapped to the slot’s [+front]/[+back] feature whether or not a liquid is adjacent. The cross-branch reflexes of these positions cluster with the divergence profile of inherited zero-grade slots, not with the stable profile of inherited real /i/ (e.g. *silā*, *piti*, *zima*; cf. the vowel control set in Section 7). The real-vowel reading would predict the latter and is therefore not consistent with the surface record. We discuss the implications further in Section 10.2.2.

9 Cross-branch extension

The validation in Sections 4 and 5 is conducted at an OCS default ranking, with results matched against conservative branch reflexes (especially South Slavic). The cross-branch divergence and pair-based profile results in Section 7 already show that branches differ systematically in how they resolve filled jer slots (Serbian /a/, Czech /e/, Russian /o/ for back jers, etc.), and that the differences pattern as partial agreement rather than random variation.

The methodology developed here generalizes naturally to a per-branch implementation. Each branch’s resolution rule (which vowel fills a filled slot) and its tolerance for syllabic resonants (whether a sonorant between obstruents wins over a filled-slot vowel) can be declared as branch-specific parameters in the same constraint framework. Implementing this and validating it against a corpus of OCS forms with attested modern reflexes across all six branches is a natural follow-up to the work reported here, but is out of scope for the present paper; we identify it as continuing work in Section 10.3.

9.1 Two demonstrations of single-parameter divergence

To illustrate that this is not idle hand-waving, we report two minimal parameter changes against the OCS-default parser that already reproduce the corresponding daughter-branch surface forms. Both demonstrations turn on the same structural principle: when the OCS configuration would leave a phonotactically dispreferred output, the slot fills to relieve the pressure.

Serbian *danās* from forbidding /dn-/ onset. OCS **d̥n̥bs̥b* ‘today’ surfaces in Czech as *dnes* and in Serbian as *danās*. The Czech reflex preserves the OCS three-jer chain with the middle ь vocalized and the outer two empty (*dnes*: /d/ onset, /n/ onset of second mora, /e/ as middle nucleus, final /s/ as coda, final ь extrametrical). The Serbian reflex vocalizes both the first and middle ь, giving /da.nas/. The difference between the two reflexes is fully accounted for by one phonotactic fact: Serbian disallows

/dn-/ as a word-initial onset cluster, whereas Czech permits it. Removing **dn** from the parser’s **legal_complex_onsets** inventory and re-running on **dɔnɔsɔ* produces mask [0, 1, 0, 1, 0, 0]—both ɛ slots fill, the chain re-anchors at the now-illegal onset, and the output is two syllables *d-a-n.a-s-(ɔ)*, recovering Serbian *danas* exactly. No other parameter changes were required.

A corroborating mini-case: Serbian has two reflexes of **sɔborɔ* ‘gathering, assembly’, *zbor* and *sabor*, both attested and both productive (the parliament of Croatia is *Hrvatski sabor*; ordinary “gathering” is *zbor*). The two reflexes differ in exactly the way the slot view predicts. The /sb-/ onset is dispreferred in Serbian. There are two phonotactic escapes: either the first ɛ fills and the cluster is broken (giving *sabor*, with the slot filled and the back-jer resolution rule producing /a/), or regressive voicing assimilation produces the licit /zb-/ onset and the slot stays empty (giving *zbor*). The same OCS form, the same Serbian phonotactic pressure, two different repair strategies, both available in the modern language. This is precisely the slot view’s expectation: when phonotactic pressure exists and an alternative repair (voicing assimilation) is also available, both outcomes can coexist as paradigm-internal or register-conditioned variants.

Russian *volk* from penalizing syllabic /l/. OCS **volkɔ* ‘wolf’ surfaces in Czech as *vlk* (syllabic /l/, slot empty) and in Russian as волк (slot filled, /l/ as coda). The Czech outcome is the OCS-default parser’s prediction: at weight 0 for **SYLLABIC-RESONANT*, /l/ becomes the nucleus at zero cost and the ɛ slot stays empty. Raising **SYLLABIC-RESONANT* from 0 to 10 (matching the weight of the heaviest constraints in the inventory, encoding “this branch does not permit syllabic resonants”) and re-running on **volkɔ* produces mask [0, 1, 0, 0, 0]—the ɛ slot fills, /l/ becomes a coda, and the single syllable parses as /v-V-lk/ with /V/ determined by Russian’s branch-specific resolution rule (front ɛ before dark /l/ surfaces as /o/, Section 1.1), giving волк exactly. No other parameter changes were required.

The general principle. Both demonstrations turn on a single structural principle: when sufficient pressure builds, the slot fills. In Serbian *danas* the pressure is phonotactic (illegal onset cluster); in Russian волк the pressure is sonority-typological (the branch does not admit syllabic /l/). The same principle has a parallel surface in modern Serbian *mrtav/mrtva* (Section 5), where a non-etymological /a/ appears in the masculine *mrtav* because the bare cluster /mrtv/ as a word-final coda is phonotactically dispreferred in Serbian. In all three cases—Serbian *danas*, Russian волк, Serbian *mrtav*—a structural slot fills when, and only when, the surrounding phonotactics demand it. The OCS-default parser handles the first by withdrawing a permission (onset inventory); it handles the second by adding a cost (syllabic-resonant penalty); and it handles the third (Section 5.3) without any parameter change at all. The unifying observation is that none

of these requires a new mechanism. The slot view’s central commitment—that filling is driven by structural pressure—scales from the OCS-internal validation corpora through the modern paradigmatic alternations to the per-branch divergence pattern.

Relation to Koenig (2004)’s dialect-specific account. Koenig (2004) analyses the *vlk*/ВОЛК divergence by appeal to dialect-specific tolerance for syllabic liquids: dialects that permitted syllabic /l/ (Czech, South Slavic) left the H-jer slot empty and made the liquid syllabic; dialects that did not (Russian, modern Polish) vocalized the H-jer instead. For the *vlk*/ВОЛК case specifically, the ВОЛК demonstration above is the mechanized version of that analysis: the single parameter *SYLLABIC-RESONANT at weight 10 encodes precisely Koenig’s “Russian does not permit syllabic /l/,” and the resulting parser output recovers ВОЛК from **vblk̥* exactly as Koenig (2004) describes.

The Serbian *danās* demonstration goes beyond what Koenig (2004) develops. There are no liquids in **d̥n̥b̥s̥v̥*; the divergence between Czech *dnes* and Serbian *danās* is driven by a phonotactic restriction on the onset cluster /dn-/, not on syllabic resonants. Koenig (2004)’s dialect-parameterization is restricted to syllabic-resonant tolerance and does not extend the same logic to onset-inventory differences. The slot view’s “slot fills under structural pressure” principle generalizes across all three sources of pressure documented here: sonority-sequencing violations (the trilemma cases, Section 6), syllabic-resonant penalties (the ВОЛК case above), and illegal onset clusters (the *danās* case above). One mechanism, three different parameter sources, three different daughter-branch outcomes.

The substantive difference between the two accounts is the unifying claim. Koenig (2004) commits to per-branch variation in syllable structure (a separate parameter per dialect for syllabic-resonant tolerance). The slot view commits to one principle (slot fills when phonotactics demand a nucleus) and allows the daughter branches to differ only in the parameter values they assign within that principle. The Czech *vlk* / Russian ВОЛК / Serbian *danās* three-way contrast is then not three separate dialect-by-dialect stories but one structural rule with three different pressure sources.

10 Discussion

10.1 Limitations

The parser is not a complete model of Slavic phonology. The major known limitations:

The three skipped Havlík forms.

**p̥b̥str̥* ‘variegated’: the word-final /str/ cluster requires either a coda inventory extension or a final-jer-as-rescue mechanism. This is a parser inventory limitation, not a theoretical issue; the slot view’s prediction (front ь vocalizes, final ь deletes, /str/ as word-final coda cluster) is the right answer, and a richer coda inventory would let the

parser produce it.

**tɒnɔkɔ* ‘thin’ and **lɒgɔkɔ* ‘light’ are worth dwelling on because they reveal a fourth class of cases, beyond those identified in Section 1.1, where standard Havlík’s Law itself produces the wrong surface prediction independent of any parser limitation. Strict Havlík counting on the flat three-jer chain proceeds from the rightmost jer: word-final ɐ is weak, the middle ɐ is followed by a weak jer so it is strong (vocalizes), and the leftmost ɐ is followed by a strong jer so it is weak (deletes). Predicted surface: **/tnVɐ/*, **/lhVɐ/*, where V is the back-jer vocalization. The modern reflexes are categorically different: Czech *tenký*, *lehký*; Serbian *tanak*, *lak*; Polish *cienki*, *lekki*; Bulgarian тънък, лек. Every branch vocalizes the leftmost ɐ, not the middle ɐ. Strict Havlík gets the wrong answer. This is a substantive failure of the counting rule, parallel to the liquid-context asymmetry and the trilemma sub-class.

The standard literature handles these forms by recognizing the derivational suffix boundary as a chain break: the root ***tɒn-* and the suffix ***-ɔkɔ* form two separate chains, each calculated independently, and within the root chain the single ɐ is strong (no jer to its right inside the chain). This gives the correct answer, but it does so by adding a morphological annotation to the chain calculation that the original Havlík formulation does not contain.

The slot view, properly stated as “syllabification operates over structural domains,” predicts exactly the morphologically-conditioned output. What the slot view requires is that the chain calculation respect morphologically defined prosodic domains, not that it operate on a flat segment string. The structural rule itself is silent on whether the chain is flat or domain-segmented; both are coherent slot-view implementations. The current parser uses the flat version because (i) it is simpler, (ii) it suffices for the validation corpora the parser was designed to test, and (iii) morphological segmentation introduces a separate set of design decisions (where suffix boundaries fall, how prefixes interact, how compounds segment) that we did not commit to from the start. The morphologically aware version is a natural follow-up: parse the segment string into morphemes, apply the syllabification rule within each morpheme, then assemble the output.

The substantive point is that the parser’s failure on **tɒnɔkɔ* and **lɒgɔkɔ* reflects an implementation choice (flat chain), not a theoretical failure of the slot view. The slot view’s general principle—that filled jer slots are determined by structural pressure within a defined domain—predicts the correct modern outputs once the domain is taken to be the morpheme. Standard Havlík has no analogous flexibility: it is a flat counting rule by stipulation, and the morphological-boundary cases are where it breaks. The two skipped forms are therefore one more empirical reason to prefer the structural reformulation over the counting reformulation.

Cross-branch implementation: demonstrated, not built out. Section 9.1 reports three single-parameter derivations against the OCS-default parser that reproduce

daughter-branch surface forms (Serbian *danas* from disallowing /dn-/; Russian *волк* from penalizing syllabic /l/; Serbian *zbor/sabor* as two phonotactic repairs of the same OCS input). The general methodology—one mechanism with branch-specific parameter values—is therefore not just speculative; the existence proofs are in the paper. What remains for follow-up work is the full per-branch implementation: declaring the complete parameter set for each of the six branches (onset and coda inventories, syllabic-resonant weight, resolution-rule mappings for filled slots) and validating the resulting per-branch parsers against a corpus of OCS forms with attested modern reflexes across all six daughter languages. That is a substantial engineering task we leave to follow-up work; the existence proofs in Section 9.1 establish that the methodology generalizes, but they do not constitute a built-out cross-branch system.

Held-out test coverage. The held-out validation comprises two strictly sealed tests totalling 60 OCS forms not in any design corpus, with predictions hand-labeled before each parser run and parameters frozen by md5 hash (Section 4.4). The combined result is 58/60; the two parser failures (**t̃ma*, **m̃zda*, both in the first sealed test) correspond to inventory gaps anticipated in advance and would be defensible additions with citation, but we do not make them in order to preserve the seal. Further extension—an even larger held-out corpus targeting forms involving iotation, prefix boundaries, or Bulgarian dialectal variation—would firm up coverage in those specific sub-cases, but the two-round sealed validation already in the paper is the principal empirical claim.

Edge cases in tokenization. OCS digraphs (/šč/, /dž/) and nasal vowels are handled as single tokens, but iotation (/j/-glides triggering palatalization) is not yet modeled.

10.2 Anticipated objections

The structural reformulation makes a strong claim—that the jers were never vocalic at the CS lexical level—and several objections to this claim are both predictable and important. We address them here in roughly the order they are likely to be raised.

10.2.1 The palatalization chronology

Objection. The three Slavic palatalizations are conventionally held to have been triggered by front vowels, including front jers Ъ. If front jers were not vocalic at the Common Slavic level, what triggered the palatalizations?

Response. The objection presumes a binary—“jers were vowels” versus “jers were not vowels”—that the slot view does not commit to. Under the slot view as developed here, jers are not simply “non-vocalic.” They are **structural slots with a [+front] or [+back] feature specification** (the slot’s “color”). Whether a given slot has surface vocalic content depends jointly on (a) the lexical type of the slot and (b) the phonotactic

conditions of the form. Different sub-classes of jer position behave differently, and this differential behavior is exactly what the palatalization pattern tracks.

Three classes of jer position

We distinguish three classes of slot, each with a different phonological function and a different relationship to palatalization triggering:

(1) *Vocalizing slots*. Slots that surface with a vocalic realization under structural pressure. These are the jers tracked by the modern nepostojano A alternation (Section 5): they fill when no other vowel anchors the syllable and stay empty when one does. When filled, these slots have actual vocalic content—a front vowel for Ъ-source, a back vowel for Ъ-source. The branch-specific resolution rules of Section 7 determine the surface quality (Serbian /a/, Czech /e/, Russian /o/, Bulgarian /ə/, etc.), but the [+front] or [+back] feature is constant. When a vocalizing Ъ-slot is filled, the resulting front vowel triggers palatalization of an adjacent consonant in exactly the same way any other front vowel does. The standard account of the First Palatalization is fully consistent with this: palatalization is triggered by the front-vowel content of filled vocalizing slots, and that content was present in the relevant historical period whenever the structural conditions for filling were met.

(2) *Connector/cluster-modifying slots*. Slots that do not surface with a vocalic realization but serve other phonological functions: marking palatalization of an adjacent consonant; participating in complex onset or coda clusters as a notational element; marking syllabicity of an adjacent sonorant (the case central to Section 6 and the Cat B argument of (Jevremović, 2026a)). The [+front] feature on these slots may still condition palatalization on the host consonant cross-linguistically, since palatalization is a featural process and does not require a vocalic carrier (Section 10.2.5 and prior phonological literature on feature-only palatalization triggers). Whether the First Palatalization was triggered by these slots specifically is an empirical question. The slot view’s prediction is that connector slots could trigger palatalization through the featural [+front] specification alone, even without vocalic surfacing.

(3) *Word-final morphological jers*. Word-final Ъ and Ъ in OCS frequently mark inflectional morphology—most prominently the masculine nominative singular suffix descending from PIE *-os, where the back jer Ъ signals the case ending in a reduced form. These positions serve a function in the inflectional paradigm; they are identifiable by the morphological category they signal, and they fall under extrametricality at the prosodic level (Section 3). Word-final jers are typically irrelevant to palatalization because they fall outside the structural conditions under which palatalization operates. They are also why our parser implements word-final jer extrametricality as a structural default: their function is morphological rather than phonological, and they are diagnosable from the inflectional paradigm without recourse to surface vocalization data.

The classification predicts the palatalization pattern

The three-way classification yields the expected pattern of palatalization triggering:

- Class (1) slots filled with front vowels trigger palatalization regularly, indistinguishable from real front vowels in the relevant contexts.
- Class (2) slots may trigger palatalization through the [+front] feature alone (a featural process), or may not, depending on whether the language treats featural palatalization as a productive process at the relevant historical stage.
- Class (3) slots typically do not trigger palatalization, since they are outside the structural domain of the process.

Standard accounts of the First Palatalization focus precisely on contexts where what we are calling class (1) slots are filled in the relevant historical period—they do not need to invoke a uniform “all jers were vowels” claim. The pattern can be derived from the slot view’s sub-classification together with the prediction that featural palatalization may have operated on class (2) slots in some cases.

Chronological dimension

The chronological question reduces to: which slots had filled-vowel content at the time of the First Palatalization, and which were already empty? Standard accounts hold that the filled state of class (1) slots was more widespread in Pre-OCS and that the modern alternation pattern (filled in some forms, empty in others) emerged as the system was reanalyzed over the following centuries. This is fully consistent with the slot view: the slot’s identity (color and class) was constant; the regularity of when class (1) slots filled changed over time.

The strong claim about pre-OCS phonetic content is not essential for the slot view. What *is* essential is the structural classification at the time of the alternation pattern that became Havlík’s Law (approximately the 9th century onward), the orthographic distribution at the time of OCS attestation, and the cross-context asymmetry within the same branch (Section 2.3). All three are decidable without committing to a categorical position on the First Palatalization triggers, and all three are independent of the particular chronology assumed for the phonetic realization of class (1) slots.

Two further points consistent with standard accounts

Not all palatalizations are jer-triggered. The Second Palatalization ($k \rightarrow c$, $g \rightarrow z$, $x \rightarrow s$) is conventionally triggered by secondary *i from *ai/*oi, which became *ě (yat)—a real vowel, not a jer (Shevelov, 1964; Carlton, 1991). The Third Palatalization is progressive, conditioned by a preceding front vowel context. Of the three palatalizations, only the

First is straightforwardly attributed to triggering by both real front vowels and class (1) front jers in standard accounts.

Palatalization is featurally conditioned across languages. Cross-linguistically, palatalization is triggered by [+front] or [-back] specifications on adjacent positions, not by vocalic content per se. This is the basis for the prediction in class (2) above: featural triggering does not require vocalic surfacing.

10.2.2 Greek loanwords spelled with jer

Objection. Greek loanwords containing iota (the front vowel /i/) were sometimes adapted into OCS with Ѣ in the orthography. If Ѣ is a slot rather than a vowel, why would scribes use it to render a Greek vowel?

Response. The pattern is the same one documented in Section 8. A Greek source syllable with iota enters Slavic and is reanalyzed as a zero-grade structural position carrying a [+front] feature, not preserved as a real /i/. The vocalic content of the source is discarded; only the featural specification is retained on the slot. The branch-specific surface vowel is then computed from the slot, not inherited from Greek. This is the same cross-source convergence pattern exhibited by **dǣlgō* (Germanic /u/ reanalyzed as a [+back] slot, Section 8.2); the Greek-iota case extends the convergence to a front-vowel source.

The evidence that the resulting position is a slot rather than a preserved /i/ comes from the divergence test of Section 7: the cross-branch surface outcomes of these loanword positions cluster with the divergence profile of inherited zero-grade slots, not with the profile of inherited real /i/. Real /i/ is stable across the six branches (the vowel control set in Section 7 contains stable-/i/ words such as *sila*, *piti*, *zima* with uniform reflexes; cf. the much lower lability mean for the vowel-control set than for the jer set). A Greek iota preserved as a real Slavic /i/ should pattern with that control. The actual reflexes pattern instead with the high-divergence jer profile, which is what the slot view predicts: the orthographic Ѣ is the [+front] slot mark, not a phonetic transcription.

Two further points reinforce the reading. First, OCS loanword orthography is conservative and follows the existing native graphemic system; there was no separate grapheme for “borrowed /i/ that should be distinguished from a native jer slot”—the available representation for a [+front] nuclear position without an inherent vocalic specification is Ѣ, and that is what scribes used. Second, the orthographic distributional restriction noted in Section 6 (jers never word-initial) applies to Greek loanwords as well: a Greek word beginning with iota is not rendered with initial Ѣ. The Greek-loanword spellings are consistent with the slot view’s treatment of Ѣ as a featural slot mark used in exactly the same distributional positions as native jers.

10.2.3 Within-branch consistency

Objection. If jer vocalization is determined by branch-specific resolution rules, why is the resolution so consistent within a single branch? Russian back jers consistently surface as /o/, Czech back jers consistently as /e/. This consistency seems easier to explain if the jers were real underlying vowels modified by regular sound change than if they were slots filled independently per word.

Response. The slot view does not predict per-word variability of filling within a branch. It predicts that each branch has a *rule* for filling each slot type, and the rule applies uniformly to all slots of that type in the branch. Within-branch consistency is exactly what the slot view predicts under a deterministic resolution rule. Cross-branch divergence (Section 7) reflects parametric variation in which rule each branch chose; within-branch consistency reflects the deterministic application of the chosen rule. The two predictions are complementary, not in tension.

The single-parameter demonstrations of Section 9.1 make this concrete: changing one phonotactic parameter against the OCS-default parser (the *SYLLABIC-RESONANT weight, or the legality of the /dn-/ onset cluster) produces a new output that applies uniformly to every form in the lexicon that meets the structural conditions. The mechanism that produces Russian волк from *vɔlkɔ is the same mechanism that would produce Russian полный from *pɔlnɔ, Russian полк from *pɔlkɔ, and so on; the within-branch consistency falls out of the architecture, not as a separate stipulation.

10.2.4 Why not abstract underlying vowels with deletion?

Objection. The classical generative analysis posits abstract underlying jers as real vowels deleted by rule when not in the right structural context. This account also predicts the surface alternation, without making the strong claim that the jers were never vocalic. Why prefer the slot view?

Response. As noted in Section 2.2, the classical generative account and the slot view make largely overlapping surface predictions for individual alternations. Scheer (2005) argues that the abstract underlying vowel, once committed to the position of being deletable by structural conditioning, is no longer phonologically distinct from an empty nucleus: “a ‘vocalic zero’ can be nothing else than an empty nucleus.” We adopt this position substantively. The slot view is therefore not in conflict with the abstract-vowel account but a terminological clarification of it: what the abstract-vowel account calls an underlying jer that always deletes outside of specific contexts is, structurally, an empty nucleus that fills only when those contexts demand a nucleus.

Where the slot view diverges substantively is in the cross-branch divergence prediction (Section 7). The abstract-vowel account, in its strongest form, posits two underlying jer vowels at CS (ɤ as /i/-like, ɐ as /u/-like or schwa-like). The comparative method

should recover these proto-qualities from daughter reflexes. The cross-branch divergence we document is hard to reconcile with this expectation. The slot view, on which the filling vowel is determined per branch rather than recovered from a proto-quality, predicts the divergence directly.

Koenig (2004) develops a sophisticated intermediate position that is worth singling out, since it represents the most direct contemporary alternative to the position adopted here. Koenig (2004) treats jers as configurations with both a skeletal X-slot and a melodic feature tier present in the underlying representation, but unlinked. Vocalization (YVoc) links the two when an unsyllabified consonant requires nuclear support. The skeletal slot is invisible to syllabification until linking. This account preserves the empty-nucleus insight at the skeletal level while keeping pre-specified vocalic content on the melodic tier. For within-branch alternations the predictions overlap with the slot view, and the directional R-to-L parsing converges. The decisive point of divergence is the cross-branch test of Section 7: because the melodic features are pre-specified at CS in Koenig (2004)’s account, the comparative method should recover a proto-quality, and surface divergence across branches must be derived through branch-specific lowering, raising, or backing of that proto-quality. The categorical divergence we document (the jer set’s near-disjoint partial-agreement profile across six branches) is more parsimoniously derived under the slot view, where there is no proto-quality to be eroded; the slot fills with whatever each branch’s resolution rule selects.

10.2.5 Bulgarian back jer as /ə/

Objection. Bulgarian retains a phonetic schwa-like reflex of the back jer ъ (modern Bulgarian ъ pronounced as [ə]). Does this not suggest that the back jer was already vocalic at CS, with Bulgarian preserving the original quality?

Response. Modern Bulgarian ъ is the Bulgarian branch’s resolution rule for the back slot: fill with [ə]. This is structurally no different from Serbian’s resolution rule of filling with /a/, Czech’s with /e/, or Russian’s with /o/. The Bulgarian rule happens to produce a vowel quality that orthographically and phonetically resembles a schwa, but the mechanism is the same: per-branch parametric resolution. The Bulgarian case is not evidence for an underlying vocalic content; it is one more data point in the parametric variation across branches.

10.3 Future work

1. **Extend the parser to handle morphological-boundary cases** (**tɒnɔkɔ*, **lɔgɔkɔ*, and similar 3-jer chains with derivational suffixes). Standard Havlík’s law conflicts with the modern data in these cases: Czech *tenký*, *lehký* vocalize the leftmost jer rather than the medial jer that strict counting would predict. These

cases were skipped from the present analysis (Section 10.1), but are arguably the cases where a structural alternative to Havlík should prove itself most clearly. Extending the parser with morphologically aware syllabification (treating the suffix *-ǫk-* as a separate prosodic domain, for instance) is a natural and high-priority extension.

2. Survey OCS for additional forms in the trilemma sub-class (Section 6). The predicted class is small but well-defined; exhaustive coverage will firm up the empirical claim.

10.4 What this paper does and does not establish

We have argued that Havlík’s Law is not a counting rule but the surface signature of a syllabification mechanism. The structural rule, mechanized as a constraint-based syllabifier, produces:

- The Havlík strong/weak distribution at the OCS level, including three-*jer* chains in one pass (52 of 52 in-scope forms).
- The Category A vs. Category B asymmetry documented in (Jevremović, 2026a), as a direct consequence of constraint ranking.
- The synchronic *nepostojano* A paradigmatic alternation in modern Serbian (31 of 31 forms across 19 lexemes, with 13 *nepostojano* A forms showing alternation as predicted and 18 real-vowel control forms showing stability as predicted), demonstrating that the same rule operates at both diachronic and synchronic levels and does not overgenerate on stable-vowel paradigms.
- A previously undocumented sub-class of word-initial syllabic resonants preserved in Serbian *rđa*, *rvati*, *mrknuti*, as a structural prediction the parser produced unprompted.
- **The existence of syllabic liquids as a class in OCS, derived without presupposition.** The trilemma argument (Section 6) shows that any analyst working with the standard tools—Havlík’s counting rule, the orthographic record, the inventory of OCS vowels—is forced into one of three options for **rǫděti*-type forms, and two of the three are independently untenable. The remaining option requires accepting that syllabic liquids were a real class in OCS phonology. This conclusion does not depend on the slot view, on the constraint-based syllabifier, or on the cross-branch divergence data; it follows from standard assumptions alone. The slot view provides the cleanest representational account of the class, but its existence is established by the trilemma without circularity.

- A statistical signature in the cross-branch data (Section 7): on a corpus of 29 jer-vocalization positions and 30 real-vowel controls in six Slavic branches, jer-source positions show categorically more cross-branch vowel quality divergence (Mann-Whitney $U = 44$, $z = -5.921$, $p < 10^{-6}$, rank-biserial effect = 0.899) than real-vowel positions, consistent with the slot view’s prediction that the filling vowel is determined per-branch rather than by a single Common Slavic proto-quality.
- A complementary argument from loanword reanalysis (Section 8): categorically different historical sources converge on the same Slavic slot structure, with the slot’s [+front]/[+back] specification retaining a featural trace of the source vowel and the vocalic content itself being absorbed into the structural pattern.
- **Two single-parameter demonstrations that the OCS-default parser plus a minimal phonotactic change reproduces daughter-branch surface forms** (Section 9.1): removing /dn-/ from the onset inventory yields Serbian *danas* from **dn̥n̥sb̥*; raising the syllabic-resonant penalty yields Russian волк from **v̥lk̥z̥*. The Czech *vlk* outcome is what the OCS-default parser produces without any change. The three surface forms—Czech *vlk*, Russian волк, Serbian *danas*—are connected by one principle: the slot fills when the surrounding phonotactics build sufficient pressure. This is a substantive cross-branch prediction the counting rule cannot make: standard Havlík operates on a flat string with no representational means to record branch-specific phonotactic parameters.

We have *not* established that the structural rule is the unique correct analysis. For many forms, an abstract-underlying-vowel reading with appropriate deletion rules makes the same surface predictions. The structural rule is preferable on grounds of (a) parsimony (one mechanism rather than two), (b) automatic coverage of the three empirical problems identified in Section 1.1, and (c) the trilemma argument (Section 6), which the alternative reading cannot accommodate without smuggling in the structural analysis.

The theoretical framework adopted—empty-nucleus syllabification with ranked constraints—is not new. The framework derives from Government Phonology and CVCV phonology (Kaye, 1990; Lowenstamm, 1996; Scheer, 2004, 2005) and is adjacent to Bethin (1998). What is new is the unification: Havlík’s Law, modern Serbian nepostojano A, and the word-initial syllabic-resonant sub-class reduce to a single ranked-constraint mechanism. The mechanism is mechanized as an algorithm and validated under sealed-test methodological discipline. On two strictly sealed held-out corpora the result is 58/60. The two parser failures correspond to inventory gaps anticipated in advance. The 95/95 in-scope on the three pre-design corpora is consistency with design, not independent validation; the held-out 58/60 is the principal result. The diachronic claim is restricted: the jers were not vocalic at the time of the alternation pattern that

became Havlík’s Law; earlier phonetic content is a separate question (Section 10.2.1). A further consequence of the slot view, with implications for the standard periodization of the so-called “fall of the jers,” requires manuscript-level engagement with the OCS attestational record and is deferred to a separate paper in preparation. Slots, not vowels; Havlík’s Law is what those slots do when forced to surface.

References

- Aleksandar Belić. *Istorija srpskohrvatskog jezika, knj. II, sv. 1: Reči sa deklinacijom*. Naučna knjiga, Beograd, 1962. Classical Serbian historical-morphological treatment of the declensional system; nepostojano A is treated as descended from the historical semivowel (poluglasnik), with a three-way origin classification (etymological jer; non-etymological development near word-final r, l, m, v; analogical extension).
- Christina Y. Bethin. *Slavic Prosody: Language Change and Phonological Theory*. Cambridge Studies in Linguistics 86. Cambridge University Press, Cambridge, 1998.
- Terence R. Carlton. *Introduction to the Phonological History of the Slavic Languages*. Slavica, Columbus, OH, 1991.
- Jacob Cohen. *Statistical Power Analysis for the Behavioral Sciences*. Lawrence Erlbaum, Hillsdale, NJ, 2nd edition, 1988.
- Rick Derksen. *Etymological Dictionary of the Slavic Inherited Lexicon*. Leiden Indo-European Etymological Dictionary Series 4. Brill, Leiden, 2008.
- Edmund Gussmann. *Studies in Abstract Phonology*. MIT Press, Cambridge, MA, 1980.
- Antonín Havlík. K otázce jerů v starou češtině. *Listy filologické*, 16:45–51, 106–116, 248–258, 342–353, 436–445, 1889.
- Bruce Hayes. *Metrical Stress Theory: Principles and Case Studies*. University of Chicago Press, Chicago, 1995.
- Junko Itô. A prosodic theory of epenthesis. *Natural Language and Linguistic Theory*, 7:217–259, 1989.
- Marko Jevremović. A categorical split in Havlík’s law: Liquid-context jers never vocalize. LingBuzz preprint, 2026a. Available at <https://lingbuzz.net/lingbuzz/009846>.
- Marko Jevremović. Back jer as a negative marker: An etymological diagnostic from slavic liquid contexts. LingBuzz preprint, 2026b. Available at <https://ling.auf.net/lingbuzz/010011>.

- Jonathan Kaye. ‘Coda’ licensing. *Phonology*, 7(2):301–330, 1990.
- Ivan Klajn. *Gramatika srpskog jezika*. Zavod za udžbenike i nastavna sredstva, Beograd, 2005.
- Jeanette Suzanne Koenig. *A Diachronic Analysis of the Interaction of Syllabification and Jer Vocalization*. PhD thesis, Princeton University, Princeton, NJ, 2004.
- Miroslav Komárek. *Historická mluvnice česká, I: Hláskosloví*. Státní pedagogické nakladatelství, Prague, 3rd edition, 1969.
- Frederik Kortlandt. The rise and development of Slavic accentual paradigms. *Studies in Baltic and Slavic Linguistics*, 2009. Working paper, Leiden University.
- Arnošt Lamprecht. *Praslovanština*. Univerzita J. E. Purkyně, Brno, 1987.
- Theodore M. Lightner. On the phonology of Old Church Slavonic conjugation. *International Journal of Slavic Linguistics and Poetics*, 9:1–28, 1965.
- Jean Lowenstamm. CV as the only syllable type. In Jacques Durand and Bernard Laks, editors, *Current Trends in Phonology: Models and Methods*, pages 419–441. European Studies Research Institute, Salford, 1996.
- Horace G. Lunt. *Old Church Slavonic Grammar*. Mouton de Gruyter, Berlin, 7th edition, 2001.
- Saskia Pronk-Tiethoff. *The Germanic Loanwords in Proto-Slavic*. Rodopi, Amsterdam, 2013.
- Jerzy Rubach. *Cyclic and Lexical Phonology: The Structure of Polish*. Foris, Dordrecht, 1984.
- Jerzy Rubach. Abstract vowels in three-dimensional phonology: the yers. *The Linguistic Review*, 5:247–280, 1986.
- Tobias Scheer. *A Lateral Theory of Phonology: What is CVCV and Why Should It Be?*, volume 1. Mouton de Gruyter, Berlin, 2004.
- Tobias Scheer. Slavic vowel-zero alternations and Government Phonology: Two approaches, one solution. In Steven Franks, Frank Gladney, and Mila Tasseva-Kurkchieva, editors, *Formal Approaches to Slavic Linguistics: The South Carolina Meeting*, pages 300–311. Michigan Slavic Publications, Ann Arbor, 2005.
- George Y. Shevelov. *A Prehistory of Slavic: The Historical Phonology of Common Slavic*. Carl Winter, Heidelberg, 1964.

Živojin Stanojčić and Ljubomir Popović. *Gramatika srpskoga jezika: udžbenik za I, II, III i IV razred srednje škole*. Zavod za udžbenike i nastavna sredstva, Beograd, 1992. Standard secondary-school grammar; nepostojano A introduced in the morphology section.

Charles E. Townsend and Laura A. Janda. *Common and Comparative Slavic: Phonology and Inflection*. Slavica, Columbus, OH, 1996.

André Vaillant. *Manuel du vieux slave*. Institut d'études slaves, Paris, 1948.